

Unit 14. Second Differentials and the Linear Model

1 Second-order theory

1.1 Second differentials and Hessians

Lemma 14.1 (Continuous partial derivatives imply differentiability). Let $\mathcal{U} \subseteq \mathbb{R}^p$ be open. If $\mathbf{F} = (F_1, \dots, F_q)' : \mathcal{U} \rightarrow \mathbb{R}^q$ has continuous first partial derivatives, then $\mathbf{F}$ is differentiable in the sense of the first differential introduced previously, and

$$d\mathbf{F}_{\mathbf{x}}[\mathbf{h}] = D\mathbf{F}(\mathbf{x})\mathbf{h}, \quad (D\mathbf{F})_{aj}(\mathbf{x}) = \frac{\partial F_a}{\partial x_j}(\mathbf{x}).$$

Origin: Course formulation from H §15.1c; MC Eqs. (7)–(11)

Proof. Fix $\mathbf{x} \in \mathcal{U}$. Choose $\rho > 0$ such that $\|\mathbf{z} - \mathbf{x}\|_2 < \rho$ implies $\mathbf{z} \in \mathcal{U}$. For $\|\mathbf{h}\|_2 < \rho$, put

$$\mathbf{x}^{(j)} = \mathbf{x} + \sum_{k=1}^j h_k \mathbf{e}_k \quad (j = 0, \dots, p).$$

For $j = 1, \dots, p$ and $0 \leq t \leq 1$,

$$\|\mathbf{x}^{(j-1)} + th_j\mathbf{e}_j - \mathbf{x}\|_2 \leq \|\mathbf{h}\|_2 < \rho,$$

so these points belong to $\mathcal{U}$. For $a = 1, \dots, q$ and $j = 1, \dots, p$, if $h_j \neq 0$, the one-variable mean-value theorem gives $0 < \theta_{aj} < 1$; if $h_j = 0$, put $\theta_{aj} = 0$. With

$$\boldsymbol{\xi}_{aj} = \mathbf{x}^{(j-1)} + \theta_{aj}h_j\mathbf{e}_j,$$

both cases satisfy

$$F_a(\mathbf{x} + \mathbf{h}) - F_a(\mathbf{x}) = \sum_{j=1}^p \frac{\partial F_a}{\partial x_j}(\boldsymbol{\xi}_{aj})h_j.$$

For $\epsilon > 0$, continuity of the finitely many partial derivatives gives $0 < \eta < \rho$ such that

$$\|\mathbf{z} - \mathbf{x}\|_2 < \eta \implies \left| \frac{\partial F_a}{\partial x_j}(\mathbf{z}) - \frac{\partial F_a}{\partial x_j}(\mathbf{x}) \right| < \frac{\epsilon}{\sqrt{pq}} \quad (a = 1, \dots, q; j = 1, \dots, p).$$

If $0 < \|\mathbf{h}\|_2 < \eta$, then

$$R_a(\mathbf{h}) := F_a(\mathbf{x} + \mathbf{h}) - F_a(\mathbf{x}) - \sum_{j=1}^p \frac{\partial F_a}{\partial x_j}(\mathbf{x}) h_j,$$

$$|R_a(\mathbf{h})| \leq \frac{\epsilon}{\sqrt{pq}} \sum_{j=1}^p |h_j| \leq \frac{\epsilon}{\sqrt{q}} \|\mathbf{h}\|_2,$$

and hence

$$\|\mathbf{R}(\mathbf{h})\|_2 < \epsilon \|\mathbf{h}\|_2.$$

The displayed linear term is therefore the first differential. □

Definition 14.2 (Second differential and Hessian). Let $\mathcal{U} \subseteq \mathbb{R}^p$ be open and let $f : \mathcal{U} \rightarrow \mathbb{R}$ have continuous second partial derivatives. The column of first partial derivatives is

$$\mathbf{p}_f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_p}(\mathbf{x}) \right)'.$$

The **Hessian** is

$$\mathbf{H}f(\mathbf{x}) = \left(\frac{\partial}{\partial x_j} \left\{ \frac{\partial f}{\partial x_i} \right\} (\mathbf{x}) \right)_{i,j=1}^p.$$

Applying Lemma 14.1 to $\mathbf{p}_f$ gives

$$\mathrm{d}(\mathbf{p}_f)_{\mathbf{x}}[\mathbf{k}] = \mathbf{H}f(\mathbf{x})\mathbf{k}.$$

Thus $\mathbf{p}_f$ is continuous by the earlier differentiability–continuity result. Lemma 14.1 applied to f , followed by the earlier first-identification result, gives

$$\mathrm{d}f_{\mathbf{x}}[\mathbf{h}] = \mathbf{p}_f(\mathbf{x})'\mathbf{h}, \quad \nabla f(\mathbf{x}) = \mathbf{p}_f(\mathbf{x}).$$

For increments $\mathbf{h}, \mathbf{k} \in \mathbb{R}^p$, define the **second differential** by

$$\mathrm{d}^2 f_{\mathbf{x}}[\mathbf{h}, \mathbf{k}] := \mathbf{h}' \mathrm{d}(\mathbf{p}_f)_{\mathbf{x}}[\mathbf{k}] = \sum_{i=1}^p \sum_{j=1}^p h_i k_j \frac{\partial}{\partial x_j} \left\{ \frac{\partial f}{\partial x_i} \right\} (\mathbf{x}).$$

For a repeated increment, write

$$d^2 f_{\mathbf{x}}[\mathbf{h}] = d^2 f_{\mathbf{x}}[\mathbf{h}, \mathbf{h}] = \mathbf{h}' H f(\mathbf{x}) \mathbf{h}.$$

Equivalently, with $d\mathbf{x} = \mathbf{h}$, $d^2 f = (d\mathbf{x})' H f(\mathbf{x}) d\mathbf{x}$.

Origin: Course formulation from MC §9; H §15.9

Theorem 14.3 (Symmetry and second identification). Under Definition 14.2, $H f(\mathbf{x})$ is symmetric. If $\mathbf{B}(\mathbf{x}) \in \mathbb{R}^{p \times p}$ satisfies

$$d^2 f_{\mathbf{x}}[\mathbf{h}] = \mathbf{h}' \mathbf{B}(\mathbf{x}) \mathbf{h} \quad (\mathbf{h} \in \mathbb{R}^p),$$

then

$$H f(\mathbf{x}) = \frac{\mathbf{B}(\mathbf{x}) + \mathbf{B}(\mathbf{x})'}{2}.$$

Conversely, this matrix identity gives the displayed second differential.

Origin: Adapted from MC Prop. 13; A&M Ex. 13.47(c)

Proof. Fix i, j and set $F(s, t) = f(\mathbf{x} + s\mathbf{e}_i + t\mathbf{e}_j)$. For sufficiently small $s, t \neq 0$, the corresponding rectangle lies in $\mathcal{U}$. Applying the one-variable mean-value theorem in the two possible orders gives points (ξ, η) and $(\tilde{\xi}, \tilde{\eta})$ in that rectangle such that

$$\frac{F(s, t) - F(s, 0) - F(0, t) + F(0, 0)}{st} = F_{st}(\xi, \eta) = F_{ts}(\tilde{\xi}, \tilde{\eta}).$$

Continuity and $(s, t) \rightarrow (0, 0)$ yield

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}).$$

Thus $\mathbf{H}f(\mathbf{x}) = \mathbf{H}f(\mathbf{x})'$.

Set

$$\mathbf{C} = \mathbf{H}f(\mathbf{x}) - \frac{\mathbf{B}(\mathbf{x}) + \mathbf{B}(\mathbf{x})'}{2}.$$

Then $\mathbf{C} = \mathbf{C}'$ and $\mathbf{h}'\mathbf{C}\mathbf{h} = 0$ for every $\mathbf{h}$. Taking $\mathbf{h} = \mathbf{e}_i$ and $\mathbf{h} = \mathbf{e}_i + \mathbf{e}_j$ gives

$$c_{ii} = 0, \quad 0 = c_{ii} + 2c_{ij} + c_{jj} = 2c_{ij},$$

so $\mathbf{C} = \mathbf{O}$. Conversely, if $\mathbf{H}f = (\mathbf{B} + \mathbf{B}')/2$, then

$$\mathbf{h}'\mathbf{H}f\mathbf{h} = \mathbf{h}'\mathbf{B}\mathbf{h} + \frac{1}{2}\mathbf{h}'(\mathbf{B}' - \mathbf{B})\mathbf{h} = \mathbf{h}'\mathbf{B}\mathbf{h}.$$

□

Theorem 14.4 (Trace identification of a matrix Hessian). Let $\mathcal{U} \subseteq \mathbb{R}^{m \times n}$ be open, let $f : \mathcal{U} \rightarrow \mathbb{R}$ have continuous second partial derivatives, fix $\mathbf{X} \in \mathcal{U}$, and let $\mathbf{x} = \text{vec } \mathbf{X}$. If $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\mathbf{B} \in \mathbb{R}^{m \times m}$, then

$$\left[d^2 f_{\mathbf{X}}[\Delta] = \text{tr}\{\mathbf{A}\Delta'\mathbf{B}\Delta\} \text{ for every } \Delta \in \mathbb{R}^{m \times n} \right] \iff \mathbf{H}_{\mathbf{x}}f(\mathbf{X}) = \frac{\mathbf{A}' \otimes \mathbf{B} + \mathbf{A} \otimes \mathbf{B}'}{2}.$$

In the square case $n = m$, put

$$\mathcal{U}_s = \{\text{vech } \mathbf{X} : \mathbf{X} = \mathbf{X}', \mathbf{X} \in \mathcal{U}\}, \quad f_s(\text{vech } \mathbf{X}) = f(\mathbf{X}).$$

Then $\mathcal{U}_s$ is open. If $\mathbf{x}_s = \text{vech } \mathbf{X}$, then, for every $\Delta = \Delta'$,

$$d\mathbf{x}_s = \text{vech } \Delta, \quad d^2(f_s)_{\mathbf{x}_s}[\text{vech } \Delta] = (d\mathbf{x}_s)'\mathbf{D}'_m \mathbf{H}_{\text{vec } \mathbf{X}}f(\mathbf{X})\mathbf{D}_m d\mathbf{x}_s.$$

Consequently,

$$\mathbf{H}f_s(\mathbf{x}_s) = \mathbf{D}'_m \mathbf{H}_{\text{vec } \mathbf{X}} f(\mathbf{X}) \mathbf{D}_m.$$

Origin: Course formulation from MC Prop. 15; A&M Ex. 13.47(c)

Proof. The vec–Kronecker identity gives

$$\text{tr}\{\mathbf{A}\mathbf{\Delta}'\mathbf{B}\mathbf{\Delta}\} = (\text{vec } \mathbf{\Delta})'(\mathbf{A}' \otimes \mathbf{B}) \text{vec } \mathbf{\Delta}.$$

Theorem 14.3 gives the first assertion. Fix a symmetric $\mathbf{X} \in \mathcal{U}$. Choose $\rho > 0$ such that $\|\mathbf{\Delta}\|_F < \rho$ implies $\mathbf{X} + \mathbf{\Delta} \in \mathcal{U}$. Then for symmetric $\mathbf{\Delta}$,

$$\|\text{vech } \mathbf{\Delta}\|_2 < \frac{\rho}{\sqrt{2}} \implies \|\mathbf{\Delta}\|_F \leq \sqrt{2} \|\text{vech } \mathbf{\Delta}\|_2 < \rho,$$

so $\mathcal{U}_s$ is open.

For symmetric increments,

$$\text{vec } \mathbf{\Delta} = \mathbf{D}_m \text{vech } \mathbf{\Delta} = \mathbf{D}_m d\mathbf{x}_s,$$

and the first-order chain rule gives

$$\nabla f_s(\mathbf{x}_s) = \mathbf{D}'_m \nabla_{\text{vec } \mathbf{X}} f(\mathbf{X}).$$

Differentiating this identity gives the stated second differential and Hessian. □

1.2 The second-order chain rule

Theorem 14.5 (Second-order chain rule). Let $\mathcal{U} \subseteq \mathbb{R}^p$ and $\mathcal{V} \subseteq \mathbb{R}^q$ be open. Let $\mathbf{g} : \mathcal{U} \rightarrow \mathcal{V}$ and $f : \mathcal{V} \rightarrow \mathbb{R}$ have continuous second partial derivatives. Write

$$\mathbf{J}(\mathbf{x}) = D\mathbf{g}(\mathbf{x}), \quad f_a(\mathbf{y}) = \frac{\partial f}{\partial y_a}(\mathbf{y}).$$

Then

$$H(f \circ \mathbf{g})(\mathbf{x}) = \mathbf{J}(\mathbf{x})' Hf(\mathbf{g}(\mathbf{x})) \mathbf{J}(\mathbf{x}) + \sum_{a=1}^q f_a(\mathbf{g}(\mathbf{x})) Hg_a(\mathbf{x}).$$

Equivalently,

$$d^2(f \circ \mathbf{g}) = (d\mathbf{g})' Hf(\mathbf{g}) d\mathbf{g} + \sum_{a=1}^q f_a(\mathbf{g}) d^2 g_a.$$

Origin: Adapted from MC Prop. 14

Proof. Lemma 14.1 and the earlier first-order chain and product rules give, for $i, j = 1, \dots, p$,

$$\begin{aligned}\frac{\partial(f \circ \mathbf{g})}{\partial x_i} &= \sum_{a=1}^q f_a(\mathbf{g}) \frac{\partial g_a}{\partial x_i}, \\ \frac{\partial^2(f \circ \mathbf{g})}{\partial x_j \partial x_i} &= \sum_{a=1}^q \sum_{b=1}^q \frac{\partial^2 f}{\partial y_b \partial y_a}(\mathbf{g}) \frac{\partial g_b}{\partial x_j} \frac{\partial g_a}{\partial x_i} + \sum_{a=1}^q f_a(\mathbf{g}) \frac{\partial^2 g_a}{\partial x_j \partial x_i}.\end{aligned}$$

These entries give the Hessian identity. Multiplication on the left and right by $d\mathbf{x}'$ and $d\mathbf{x}$, respectively, gives the second differential identity. $\square$

Example 14.1 (Weighted least-squares Hessian). Let $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times p}$, and $\mathbf{W} = \mathbf{W}' \in \mathbb{R}^{n \times n}$ with $\mathbf{W} \succeq \mathbf{0}$. Define

$$q(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' \mathbf{W} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

For $\mathbf{r} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$,

$$dq = -2\mathbf{r}' \mathbf{W} \mathbf{X} d\boldsymbol{\beta}, \quad d^2q = 2(d\boldsymbol{\beta})' \mathbf{X}' \mathbf{W} \mathbf{X} d\boldsymbol{\beta},$$

and therefore

$$\nabla q(\beta) = -2\mathbf{X}'\mathbf{W}\mathbf{r}, \quad \mathbf{H}q(\beta) = 2\mathbf{X}'\mathbf{W}\mathbf{X}.$$

If $\mathbf{W} \succ \mathbf{0}$ and $\mathbf{X}$ has full column rank, then $\mathbf{H}q(\beta) \succ \mathbf{0}$.

Origin: Adapted from A&M Exs. 13.54–13.55; H Thm. 14.12.20

1.3 Convexity and stationary points

Definition 14.6 (Convexity and stationarity). A set $\mathcal{C} \subseteq \mathbb{R}^p$ is **convex** if

$$(1-t)\mathbf{x} + t\mathbf{y} \in \mathcal{C} \quad (\mathbf{x}, \mathbf{y} \in \mathcal{C}, 0 \leq t \leq 1).$$

For a convex set $\mathcal{C}$, a function $f : \mathcal{C} \rightarrow \mathbb{R}$ is **convex** if

$$f((1-t)\mathbf{x} + t\mathbf{y}) \leq (1-t)f(\mathbf{x}) + tf(\mathbf{y});$$

it is **strictly convex** when the inequality is strict for $\mathbf{x} \neq \mathbf{y}$ and $0 < t < 1$. If f is differentiable at an interior point $\mathbf{x}_*$, it is **stationary** when

$$df_{\mathbf{x}_*}[\mathbf{h}] = 0 \quad (\mathbf{h} \in \mathbb{R}^p), \quad \text{equivalently} \quad \nabla f(\mathbf{x}_*) = \mathbf{0}.$$

A point $\mathbf{x}_*$ is a **global minimiser** when $f(\mathbf{x}_*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in \mathcal{C}$. It is a **local minimiser** when, for some $\delta > 0$,

$$\mathbf{x} \in \mathcal{C}, \|\mathbf{x} - \mathbf{x}_*\|_2 < \delta \implies f(\mathbf{x}_*) \leq f(\mathbf{x});$$

the minimum is **strict** when equality is excluded for $\mathbf{x} \neq \mathbf{x}_*$. Global and local maximisers reverse the corresponding inequalities. A **saddle point** satisfies, for every $\delta > 0$,

$$f(\mathbf{x}_-) < f(\mathbf{x}_*) < f(\mathbf{x}_+)$$

for some $\mathbf{x}_-, \mathbf{x}_+ \in \mathcal{C}$ within distance δ of $\mathbf{x}_*$.

Origin: Course formulation from MC §5; H §19.1(c)

Theorem 14.7 (Hessian criterion and stationary classification). Let $\mathcal{C} \subseteq \mathbb{R}^p$ be open and convex and let $f : \mathcal{C} \rightarrow \mathbb{R}$ have continuous second partial derivatives.

- (a) f is convex if and only if $\mathbf{H}f(\mathbf{x}) \succeq \mathbf{O}$ for every $\mathbf{x} \in \mathcal{C}$.
- (b) If $\mathbf{H}f(\mathbf{x}) \succ \mathbf{O}$ for every $\mathbf{x} \in \mathcal{C}$, then f is strictly convex.
- (c) If f is convex and $\mathbf{x}_*$ is stationary, then $\mathbf{x}_*$ is a global minimiser. If f is strictly convex, it

is unique.

- (d) At a stationary $\mathbf{x}_*$, a positive-definite Hessian gives a strict local minimum, a negative-definite Hessian gives a strict local maximum, and an indefinite Hessian gives a saddle point.

Origin: Course formulation from MC §5; H §19.1(c)

Proof. For $\mathbf{x}, \mathbf{y} \in \mathcal{C}$, set

$$\phi(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})), \quad \phi''(t) = (\mathbf{y} - \mathbf{x})' \mathbf{H} f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) (\mathbf{y} - \mathbf{x}).$$

If every Hessian is positive semidefinite, then $\phi'' \geq 0$ and the one-variable derivative ϕ' is nondecreasing; hence ϕ is convex. If every Hessian is positive definite and $\mathbf{x} \neq \mathbf{y}$, then $\phi'' > 0$, so ϕ is strictly convex.

Conversely, convexity of f makes every such ϕ convex. For $\mathbf{h} \in \mathbb{R}^p$ and sufficiently small t ,

$$f(\mathbf{x} + t\mathbf{h}) + f(\mathbf{x} - t\mathbf{h}) - 2f(\mathbf{x}) \geq 0.$$

Division by t^2 and $t \rightarrow 0$ gives $\mathbf{h}' \mathbf{H} f(\mathbf{x}) \mathbf{h} \geq 0$.

For $0 < t \leq 1$, convexity gives

$$\frac{f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{t} \leq f(\mathbf{y}) - f(\mathbf{x}).$$

Letting $t \downarrow 0$ yields $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})'(\mathbf{y} - \mathbf{x})$. At a stationary point this yields $f(\mathbf{y}) \geq f(\mathbf{x}_*)$. If a distinct $\mathbf{y}$ had equality under strict convexity, then

$$f\left(\frac{\mathbf{x}_* + \mathbf{y}}{2}\right) < \frac{f(\mathbf{x}_*) + f(\mathbf{y})}{2} = f(\mathbf{x}_*),$$

contradicting global minimality.

If $\mathbf{H}f(\mathbf{x}_*) \succ \mathbf{O}$, put $\lambda = \lambda_{\min}\{\mathbf{H}f(\mathbf{x}_*)\} > 0$. Continuity gives $\delta_1 > 0$ such that

$$|\{\mathbf{H}f(\mathbf{x}) - \mathbf{H}f(\mathbf{x}_*)\}_{ij}| < \frac{\lambda}{2p} \quad (i, j = 1, \dots, p) \quad \text{when } \|\mathbf{x} - \mathbf{x}_*\|_2 < \delta_1.$$

Hence

$$\|\mathbf{H}f(\mathbf{x}) - \mathbf{H}f(\mathbf{x}_*)\|_2 \leq \|\mathbf{H}f(\mathbf{x}) - \mathbf{H}f(\mathbf{x}_*)\|_F < \frac{\lambda}{2}.$$

Since $\mathcal{C}$ is open, choose $\rho > 0$ with $\{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_*\|_2 < \rho\} \subseteq \mathcal{C}$, and put $\delta = \min(\delta_1, \rho)$. Hence

$$\mathbf{h}'\mathbf{H}f(\mathbf{x}_* + t\mathbf{h})\mathbf{h} \geq \frac{\lambda}{2}\|\mathbf{h}\|_2^2 \quad (0 \leq t \leq 1, \ 0 < \|\mathbf{h}\|_2 < \delta).$$

Twice applying the one-variable fundamental theorem of calculus gives

$$f(\mathbf{x}_* + \mathbf{h}) - f(\mathbf{x}_*) = \int_0^1 (1-t) \mathbf{h}'\mathbf{H}f(\mathbf{x}_* + t\mathbf{h})\mathbf{h} \, dt \geq \frac{\lambda}{4}\|\mathbf{h}\|_2^2.$$

The negative-definite case applies this argument to $-f$. If the Hessian is indefinite, choose $\mathbf{u}, \mathbf{v}$ with

$$\mathbf{u}'\mathbf{H}f(\mathbf{x}_*)\mathbf{u} > 0, \quad \mathbf{v}'\mathbf{H}f(\mathbf{x}_*)\mathbf{v} < 0.$$

The same integral formula has opposite signs along sufficiently short nonzero multiples of $\mathbf{u}$ and $\mathbf{v}$. □

Example 14.2 (The log-determinant barrier). On the cone of symmetric positive-definite $m \times m$

matrices, let

$$\phi(\mathbf{\Omega}) = -\log |\mathbf{\Omega}|, \quad \boldsymbol{\omega} = \text{vech } \mathbf{\Omega}.$$

The inverse and log-determinant differential rules give

$$d\phi = -\text{tr}(\mathbf{\Omega}^{-1}d\mathbf{\Omega}),$$

$$d^2\phi = \text{tr}\{\mathbf{\Omega}^{-1}(d\mathbf{\Omega})\mathbf{\Omega}^{-1}(d\mathbf{\Omega})\} = (d\boldsymbol{\omega})'\mathbf{D}'_m(\mathbf{\Omega}^{-1} \otimes \mathbf{\Omega}^{-1})\mathbf{D}_m d\boldsymbol{\omega}.$$

For $\mathbf{0} \neq \mathbf{\Delta} = \mathbf{\Delta}'$,

$$\text{tr}\{\mathbf{\Omega}^{-1}\mathbf{\Delta}\mathbf{\Omega}^{-1}\mathbf{\Delta}\} = \text{tr}\{(\mathbf{\Omega}^{-1/2}\mathbf{\Delta}\mathbf{\Omega}^{-1/2})^2\} > 0.$$

If $\lambda = \lambda_{\min}(\mathbf{\Omega})$, then

$$\|\text{vech } \mathbf{\Delta}\|_2 < \frac{\lambda}{\sqrt{2}} \implies \|\mathbf{\Delta}\|_2 \leq \|\mathbf{\Delta}\|_F \leq \sqrt{2}\|\text{vech } \mathbf{\Delta}\|_2 < \lambda,$$

and hence

$$\mathbf{x}'(\mathbf{\Omega} + \mathbf{\Delta})\mathbf{x} \geq \{\lambda - \|\mathbf{\Delta}\|_2\}\|\mathbf{x}\|_2^2 > 0 \quad (\mathbf{x} \neq \mathbf{0}).$$

Thus the positive-definite cone is open in $\boldsymbol{\omega} = \text{vech } \boldsymbol{\Omega}$ coordinates. For $0 < t < 1$ and $\boldsymbol{\Omega}_1, \boldsymbol{\Omega}_2 \succ \mathbf{0}$,

$$\mathbf{x}'\{(1-t)\boldsymbol{\Omega}_1 + t\boldsymbol{\Omega}_2\}\mathbf{x} = (1-t)\mathbf{x}'\boldsymbol{\Omega}_1\mathbf{x} + t\mathbf{x}'\boldsymbol{\Omega}_2\mathbf{x} > 0 \quad (\mathbf{x} \neq \mathbf{0}).$$

Thus the positive-definite cone is convex, and Theorem 14.7 makes ϕ strictly convex.

Origin: Adapted from A&M Ex. 13.52; H Eq. (15.9.3); MC §11

2 Linear estimation under moment assumptions

2.1 The moment model and ordinary least squares

Definition 14.8 (Linear moment model and BLUE). Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be fixed with $\text{rank } \mathbf{X} = p$ and let $\boldsymbol{\beta} \in \mathbb{R}^p$. Let $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ be a random vector with finite second moments satisfying

$$\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}, \quad \text{cov}(\boldsymbol{\varepsilon}) = \boldsymbol{\Omega} \succ \mathbf{0}.$$

The **linear moment model** is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

For $\mathbf{C} \in \mathbb{R}^{p \times n}$, an estimator $\mathbf{C}\mathbf{y}$ is **linear**; it is **unbiased** when

$$\mathbb{E}[\mathbf{C}\mathbf{y}] = \boldsymbol{\beta} \quad (\boldsymbol{\beta} \in \mathbb{R}^p).$$

A linear unbiased estimator $\tilde{\boldsymbol{\beta}}$ is **best linear unbiased**, abbreviated BLUE, when

$$\text{cov}(\mathbf{C}\mathbf{y}) - \text{cov}(\tilde{\boldsymbol{\beta}}) \succeq \mathbf{0}$$

for every linear unbiased estimator $\mathbf{C}\mathbf{y}$ of $\boldsymbol{\beta}$.

Origin: Course formulation from A&M Ex. 13.58; CLM §§1, 3

Theorem 14.9 (Ordinary least squares and Gauss–Markov). In Definition 14.8, suppose $\boldsymbol{\Omega} = \sigma^2 \mathbf{I}_n$, where $\sigma^2 > 0$. Set

$$\mathbf{G}_0 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad \hat{\boldsymbol{\beta}}_{\text{OLS}} = \hat{\boldsymbol{\beta}} = \mathbf{G}_0\mathbf{y}.$$

The ordinary least-squares (OLS) estimator $\hat{\boldsymbol{\beta}}$ is the unique minimiser of

$$q_0(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2,$$

and the unique BLUE of $\boldsymbol{\beta}$. Moreover,

$$\mathbb{E}[\widehat{\boldsymbol{\beta}}] = \boldsymbol{\beta}, \quad \text{cov}(\widehat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

If $\mathbf{C}\mathbf{y}$ is linear and unbiased and $\mathbf{D} = \mathbf{C} - \mathbf{G}_0$, then

$$\text{cov}(\mathbf{C}\mathbf{y}) - \text{cov}(\widehat{\boldsymbol{\beta}}) = \sigma^2\mathbf{D}\mathbf{D}'.$$

Origin: Course formulation from A&M Ex. 13.57; H Thm. 19.2.2; CLM §3

Proof. Since

$$\begin{aligned} \mathbf{X}'(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}) &= \mathbf{0}, \\ q_0(\boldsymbol{\beta}) &= q_0(\widehat{\boldsymbol{\beta}}) + (\boldsymbol{\beta} - \widehat{\boldsymbol{\beta}})' \mathbf{X}'\mathbf{X}(\boldsymbol{\beta} - \widehat{\boldsymbol{\beta}}). \end{aligned}$$

For $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{a}'\mathbf{X}'\mathbf{X}\mathbf{a} = \|\mathbf{X}\mathbf{a}\|_2^2 > 0,$$

so the minimiser is unique.

The earlier affine expectation and covariance identities give

$$\mathbf{E}[\mathbf{C}\mathbf{y}] = \mathbf{C}\mathbf{X}\boldsymbol{\beta}, \quad \text{cov}(\mathbf{C}\mathbf{y}) = \sigma^2 \mathbf{C}\mathbf{C}'.$$

Thus

$$\mathbf{C}\mathbf{y} \text{ is unbiased for every } \boldsymbol{\beta} \iff \mathbf{C}\mathbf{X} = \mathbf{I}_p.$$

Since $\mathbf{G}_0\mathbf{X} = \mathbf{I}_p$, $\widehat{\boldsymbol{\beta}}$ is unbiased. Moreover,

$$\text{cov}(\widehat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{G}_0 \mathbf{G}_0' = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}.$$

For any unbiased $\mathbf{C}\mathbf{y}$,

$$\mathbf{D}\mathbf{X} = \mathbf{0}, \quad \mathbf{G}_0\mathbf{D}' = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' = \mathbf{0}.$$

Consequently,

$$\begin{aligned} \text{cov}(\mathbf{C}\mathbf{y}) - \text{cov}(\widehat{\boldsymbol{\beta}}) &= \sigma^2 \{(\mathbf{G}_0 + \mathbf{D})(\mathbf{G}_0 + \mathbf{D})' - \mathbf{G}_0\mathbf{G}_0'\} \\ &= \sigma^2 \mathbf{D}\mathbf{D}' \succeq \mathbf{0}. \end{aligned}$$

If $\mathbf{C}\mathbf{y}$ were also BLUE, comparison with $\hat{\boldsymbol{\beta}}$ would give $-\sigma^2\mathbf{D}\mathbf{D}' \succeq \mathbf{0}$. Hence

$$0 = \text{tr}(\mathbf{D}\mathbf{D}') = \sum_{i,j} d_{ij}^2, \quad \mathbf{D} = \mathbf{0},$$

so the BLUE is unique. □

2.2 Generalized least squares and Aitken's theorem

Theorem 14.10 (Generalized least squares and Aitken). Under Definition 14.8, put

$$\mathbf{Q} = \mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X}, \quad \mathbf{V} = \mathbf{Q}^{-1}, \quad \mathbf{G} = \mathbf{V}\mathbf{X}'\boldsymbol{\Omega}^{-1}.$$

Then $\mathbf{Q} \succ \mathbf{0}$, and

$$\hat{\boldsymbol{\beta}}_{\mathbf{G}} = \mathbf{G}\mathbf{y}$$

is the generalized least-squares (GLS) estimator. It is both the unique minimiser of

$$q_{\boldsymbol{\Omega}}(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\boldsymbol{\Omega}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

and the unique BLUE of $\boldsymbol{\beta}$, with

$$\mathbb{E}[\widehat{\boldsymbol{\beta}}_G] = \boldsymbol{\beta}, \quad \text{cov}(\widehat{\boldsymbol{\beta}}_G) = \mathbf{V}.$$

For every linear unbiased $\mathbf{C}\mathbf{y}$, with $\mathbf{D} = \mathbf{C} - \mathbf{G}$,

$$\text{cov}(\mathbf{C}\mathbf{y}) - \mathbf{V} = \mathbf{D}\boldsymbol{\Omega}\mathbf{D}' \succeq \mathbf{0}.$$

Origin: Course formulation from A&M Exs. 13.55, 13.58; H Thms. 14.12.19–14.12.20 and 19.2.2; CLM Ex. B5

Proof. For $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{a}'\mathbf{Q}\mathbf{a} = (\mathbf{X}\mathbf{a})'\boldsymbol{\Omega}^{-1}(\mathbf{X}\mathbf{a}) > 0,$$

so $\mathbf{Q} \succ \mathbf{0}$. The identities

$$\mathbf{G}\mathbf{X} = \mathbf{I}_p, \quad \mathbf{X}'\boldsymbol{\Omega}^{-1}(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}_G) = \mathbf{0}$$

give, for every $\boldsymbol{\beta}$,

$$q_{\boldsymbol{\Omega}}(\boldsymbol{\beta}) = q_{\boldsymbol{\Omega}}(\widehat{\boldsymbol{\beta}}_G) + (\boldsymbol{\beta} - \widehat{\boldsymbol{\beta}}_G)'\mathbf{Q}(\boldsymbol{\beta} - \widehat{\boldsymbol{\beta}}_G).$$

Thus $\widehat{\boldsymbol{\beta}}_{\mathbf{G}}$ is the unique minimiser.

The affine moment identities yield

$$\mathbf{E}[\widehat{\boldsymbol{\beta}}_{\mathbf{G}}] = \mathbf{GX}\boldsymbol{\beta} = \boldsymbol{\beta}, \quad \text{cov}(\widehat{\boldsymbol{\beta}}_{\mathbf{G}}) = \mathbf{G}\boldsymbol{\Omega}\mathbf{G}' = \mathbf{V}.$$

If $\mathbf{Cy}$ is unbiased, then $\mathbf{DX} = \mathbf{0}$ and

$$\mathbf{G}\boldsymbol{\Omega}\mathbf{D}' = \mathbf{VX}'\mathbf{D}' = \mathbf{V}(\mathbf{DX})' = \mathbf{0}.$$

Hence

$$\text{cov}(\mathbf{Cy}) = \mathbf{V} + \mathbf{D}\boldsymbol{\Omega}\mathbf{D}'.$$

If $\mathbf{Cy}$ were also BLUE, comparison with $\widehat{\boldsymbol{\beta}}_{\mathbf{G}}$ would give $-\mathbf{D}\boldsymbol{\Omega}\mathbf{D}' \succeq \mathbf{0}$; hence $\mathbf{D}\boldsymbol{\Omega}\mathbf{D}' = \mathbf{0}$. For the positive-definite square root $\boldsymbol{\Omega}^{1/2}$,

$$\mathbf{D}\boldsymbol{\Omega}\mathbf{D}' = (\mathbf{D}\boldsymbol{\Omega}^{1/2})(\mathbf{D}\boldsymbol{\Omega}^{1/2})' = \mathbf{0} \implies \mathbf{D}\boldsymbol{\Omega}^{1/2} = \mathbf{0} \implies \mathbf{D} = \mathbf{0}.$$

Thus the BLUE is unique. □

2.3 Exact linear restrictions

Theorem 14.11 (Restricted generalized least squares). In Theorem 14.10, let $1 \leq q \leq p$, let $\mathbf{R} \in \mathbb{R}^{q \times p}$ have rank q , and let $\mathbf{c} \in \mathbb{R}^q$. Suppose

$$\mathbf{R}\boldsymbol{\beta} = \mathbf{c}.$$

Define

$$\mathbf{S} = \mathbf{RVR}', \quad \mathbf{g} = \mathbf{R}\hat{\boldsymbol{\beta}}_G - \mathbf{c}.$$

Then $\mathbf{S} \succ \mathbf{O}$, and the unique constrained minimiser is

$$\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}}_G - \mathbf{VR}'\mathbf{S}^{-1}\mathbf{g}.$$

It satisfies

$$q_{\Omega}(\hat{\boldsymbol{\beta}}_R) - q_{\Omega}(\hat{\boldsymbol{\beta}}_G) = \mathbf{g}'\mathbf{S}^{-1}\mathbf{g},$$

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_R] = \boldsymbol{\beta},$$

$$\text{cov}(\hat{\boldsymbol{\beta}}_R) = \mathbf{V} - \mathbf{VR}'\mathbf{S}^{-1}\mathbf{RV}.$$

Thus restriction reduces covariance by $\mathbf{VR}'\mathbf{S}^{-1}\mathbf{RV} \succeq \mathbf{O}$.

Origin: Course formulation from A&M Ex. 13.56; H Thm. 19.2.1 and §19.7; CLM §10

Proof. For $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{a}'\mathbf{S}\mathbf{a} = (\mathbf{R}'\mathbf{a})'\mathbf{V}(\mathbf{R}'\mathbf{a}) > 0,$$

because $\mathbf{R}'$ has null space $\{\mathbf{0}\}$. Therefore $\mathbf{S}$ is positive definite. Direct substitution gives

$$\mathbf{R}\hat{\boldsymbol{\beta}}_{\text{R}} = \mathbf{R}\hat{\boldsymbol{\beta}}_{\text{G}} - \mathbf{S}\mathbf{S}^{-1}\mathbf{g} = \mathbf{c}.$$

If $\mathbf{R}\tilde{\boldsymbol{\beta}} = \mathbf{c}$, put $\mathbf{d} = \tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{\text{R}}$. Then $\mathbf{R}\mathbf{d} = \mathbf{0}$ and

$$(\hat{\boldsymbol{\beta}}_{\text{R}} - \hat{\boldsymbol{\beta}}_{\text{G}})'\mathbf{Q}\mathbf{d} = -\mathbf{g}'\mathbf{S}^{-1}\mathbf{R}\mathbf{d} = 0.$$

The quadratic decomposition in Theorem 14.10 now gives

$$q_{\Omega}(\tilde{\boldsymbol{\beta}}) = q_{\Omega}(\hat{\boldsymbol{\beta}}_{\text{R}}) + \mathbf{d}'\mathbf{Q}\mathbf{d},$$

so the constrained minimiser is unique. Taking $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}_{\text{R}}$ in the decomposition from Theorem 14.10

gives

$$\begin{aligned} q_{\Omega}(\hat{\boldsymbol{\beta}}_{\text{R}}) - q_{\Omega}(\hat{\boldsymbol{\beta}}_{\text{G}}) &= \mathbf{g}'\mathbf{S}^{-1}\mathbf{R}\mathbf{V}\mathbf{Q}\mathbf{V}\mathbf{R}'\mathbf{S}^{-1}\mathbf{g} \\ &= \mathbf{g}'\mathbf{S}^{-1}\mathbf{g}. \end{aligned}$$

Under $\mathbf{R}\boldsymbol{\beta} = \mathbf{c}$, $\mathbb{E}[\mathbf{g}] = \mathbf{0}$ and

$$\hat{\boldsymbol{\beta}}_{\text{R}} - \boldsymbol{\beta} = \{\mathbf{I}_p - \mathbf{V}\mathbf{R}'\mathbf{S}^{-1}\mathbf{R}\}(\hat{\boldsymbol{\beta}}_{\text{G}} - \boldsymbol{\beta}).$$

Using $\text{cov}(\hat{\boldsymbol{\beta}}_{\text{G}}) = \mathbf{V}$ and $\mathbf{S} = \mathbf{R}\mathbf{V}\mathbf{R}'$ gives

$$\text{cov}(\hat{\boldsymbol{\beta}}_{\text{R}}) = \mathbf{V} - \mathbf{V}\mathbf{R}'\mathbf{S}^{-1}\mathbf{R}\mathbf{V}.$$

Finally,

$$\mathbf{V}\mathbf{R}'\mathbf{S}^{-1}\mathbf{R}\mathbf{V} = (\mathbf{V}\mathbf{R}'\mathbf{S}^{-1/2})(\mathbf{V}\mathbf{R}'\mathbf{S}^{-1/2})' \succeq \mathbf{0}.$$

□

3 Gaussian criteria and sensitivity

3.1 A Gaussian criterion with unknown mean and covariance

Definition 14.12 (Gaussian criterion). For observed vectors $\mathbf{y}_1, \dots, \mathbf{y}_n \in \mathbb{R}^m$, define, for $\boldsymbol{\mu} \in \mathbb{R}^m$ and $\boldsymbol{\Omega} = \boldsymbol{\Omega}' \succ \mathbf{0}$,

$$\ell(\boldsymbol{\mu}, \boldsymbol{\Omega}) = -\frac{nm}{2} \log(2\pi) - \frac{n}{2} \log |\boldsymbol{\Omega}| - \frac{1}{2} \sum_{i=1}^n (\mathbf{y}_i - \boldsymbol{\mu})' \boldsymbol{\Omega}^{-1} (\mathbf{y}_i - \boldsymbol{\mu}).$$

A **Gaussian-criterion estimate** is a maximiser of ℓ . Put

$$\bar{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i, \quad \mathbf{S} = \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})'$$

Origin: Adapted from A&M Ex. 13.61; MC §13

Theorem 14.13 (Global Gaussian-criterion maximiser). The criterion in Definition 14.12 has a

global maximiser if and only if $\mathbf{S} \succ \mathbf{O}$. In that case the maximiser is unique and equals

$$\hat{\boldsymbol{\mu}} = \bar{\mathbf{y}}, \quad \hat{\boldsymbol{\Omega}} = \frac{1}{n}\mathbf{S}.$$

If $\mathbf{S}$ is singular, the criterion is unbounded above. For symmetric increments $d\boldsymbol{\Omega} = d\boldsymbol{\Omega}'$, put

$$\mathbf{Z}(\boldsymbol{\mu}) = \sum_{i=1}^n (\mathbf{y}_i - \boldsymbol{\mu})(\mathbf{y}_i - \boldsymbol{\mu})'.$$

Then

$$\begin{aligned} d\ell &= n(d\boldsymbol{\mu})'\boldsymbol{\Omega}^{-1}(\bar{\mathbf{y}} - \boldsymbol{\mu}) \\ &\quad + \frac{1}{2} \operatorname{tr}[\boldsymbol{\Omega}^{-1}\{\mathbf{Z}(\boldsymbol{\mu}) - n\boldsymbol{\Omega}\}\boldsymbol{\Omega}^{-1}d\boldsymbol{\Omega}] . \end{aligned}$$

Origin: Course formulation from A&M Exs. 13.61–13.64; MC §13

Proof. For every $\mathbf{x} \in \mathbb{R}^m$,

$$\mathbf{x}'\mathbf{S}\mathbf{x} = \sum_{i=1}^n \{(\mathbf{y}_i - \bar{\mathbf{y}})' \mathbf{x}\}^2 \geq 0,$$

so $\mathbf{S} = \mathbf{S}' \succeq \mathbf{O}$.

The inverse and log-determinant differential rules give

$$d\mathbf{\Omega}^{-1} = -\mathbf{\Omega}^{-1}(d\mathbf{\Omega})\mathbf{\Omega}^{-1}, \quad d \log |\mathbf{\Omega}| = \text{tr}(\mathbf{\Omega}^{-1}d\mathbf{\Omega}).$$

Also,

$$d\mathbf{Z} = -n(d\boldsymbol{\mu})(\bar{\mathbf{y}} - \boldsymbol{\mu})' - n(\bar{\mathbf{y}} - \boldsymbol{\mu})(d\boldsymbol{\mu})'.$$

Substitution in $\ell = \text{constant} - (n/2) \log |\mathbf{\Omega}| - (1/2) \text{tr}(\mathbf{\Omega}^{-1}\mathbf{Z})$ gives the stated differential.

The identity

$$\mathbf{Z}(\boldsymbol{\mu}) = \mathbf{S} + n(\boldsymbol{\mu} - \bar{\mathbf{y}})(\boldsymbol{\mu} - \bar{\mathbf{y}})'$$

implies, for each fixed $\mathbf{\Omega} \succ \mathbf{O}$,

$$\ell(\bar{\mathbf{y}}, \mathbf{\Omega}) - \ell(\boldsymbol{\mu}, \mathbf{\Omega}) = \frac{n}{2}(\boldsymbol{\mu} - \bar{\mathbf{y}})' \mathbf{\Omega}^{-1}(\boldsymbol{\mu} - \bar{\mathbf{y}}) \geq 0,$$

with equality only at $\boldsymbol{\mu} = \bar{\mathbf{y}}$.

Assume $\mathbf{S} \succ \mathbf{0}$. Set $\hat{\boldsymbol{\Omega}} = \mathbf{S}/n$ and

$$\mathbf{C} = \hat{\boldsymbol{\Omega}}^{-1/2} \boldsymbol{\Omega} \hat{\boldsymbol{\Omega}}^{-1/2} \succ \mathbf{0}.$$

If $c_1, \dots, c_m > 0$ are its eigenvalues, then

$$\begin{aligned} \ell(\bar{\mathbf{y}}, \hat{\boldsymbol{\Omega}}) - \ell(\bar{\mathbf{y}}, \boldsymbol{\Omega}) &= \frac{n}{2} \{ \log |\mathbf{C}| + \text{tr}(\mathbf{C}^{-1}) - m \} \\ &= \frac{n}{2} \sum_{j=1}^m \{ \log c_j + c_j^{-1} - 1 \}. \end{aligned}$$

For $g(t) = \log t + t^{-1} - 1$ on $t > 0$,

$$g'(t) = \frac{t-1}{t^2}, \quad g'(t) \begin{cases} < 0, & 0 < t < 1, \\ > 0, & t > 1, \end{cases}$$

so $g(t) \geq g(1) = 0$, with equality only at $t = 1$. Thus equality in the criterion comparison requires $\mathbf{C} = \mathbf{I}_m$, or $\mathbf{\Omega} = \hat{\mathbf{\Omega}}$. Consequently,

$$\ell(\boldsymbol{\mu}, \mathbf{\Omega}) \leq \ell(\bar{\mathbf{y}}, \mathbf{\Omega}) \leq \ell(\bar{\mathbf{y}}, \hat{\mathbf{\Omega}}),$$

with equality throughout exactly at $(\boldsymbol{\mu}, \mathbf{\Omega}) = (\bar{\mathbf{y}}, \hat{\mathbf{\Omega}})$.

If $\mathbf{S}$ is singular, let $r = \text{rank } \mathbf{S} < m$ and use the spectral theorem to write

$$\mathbf{S} = \mathbf{U} \text{diag}(s_1, \dots, s_r, 0, \dots, 0) \mathbf{U}', \quad s_j > 0.$$

For $t > 0$, put

$$\mathbf{\Omega}_t = \mathbf{U} \text{diag}(s_1/n, \dots, s_r/n, t, \dots, t) \mathbf{U}' \succ \mathbf{0}.$$

Then $\text{tr}(\mathbf{\Omega}_t^{-1} \mathbf{S}) = nr$ and

$$\ell(\bar{\mathbf{y}}, \mathbf{\Omega}_t) = C - \frac{n(m-r)}{2} \log t \longrightarrow +\infty \quad (t \downarrow 0),$$

where C is independent of t . □

3.2 Expected negative Hessian in vech coordinates

Definition 14.14 (Information of the Gaussian criterion). For the criterion in Definition 14.12, let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be random vectors satisfying

$$\mathbb{E}[\mathbf{y}_i] = \boldsymbol{\mu}, \quad \text{cov}(\mathbf{y}_i) = \boldsymbol{\Omega} \succ \mathbf{0} \quad (i = 1, \dots, n).$$

No cross-observation moment condition is imposed. For

$$\boldsymbol{\omega} = \text{vech } \boldsymbol{\Omega}, \quad \boldsymbol{\theta} = (\boldsymbol{\mu}', \boldsymbol{\omega}')',$$

define the **information matrix of the criterion** by

$$\mathcal{I}(\boldsymbol{\theta}) = -\mathbb{E}\{\mathbf{H}_{\boldsymbol{\theta}}\ell(\boldsymbol{\mu}, \boldsymbol{\Omega})\},$$

whenever the displayed entrywise expectations exist.

Origin: Course formulation from A&M Ex. 13.65

Theorem 14.15 (Block information formula). Under Definition 14.14,

$$\mathcal{I}(\boldsymbol{\theta}) = n \begin{pmatrix} \boldsymbol{\Omega}^{-1} & \mathbf{0} \\ \mathbf{0} & \frac{1}{2} \mathbf{D}'_m (\boldsymbol{\Omega}^{-1} \otimes \boldsymbol{\Omega}^{-1}) \mathbf{D}_m \end{pmatrix}.$$

This identity uses only the stated first and second moments.

Origin: Course formulation from A&M Ex. 13.65(a); MC Eq. (48)

Proof. For one observation, set

$$\mathbf{z} = \mathbf{y}_i - \boldsymbol{\mu}, \quad \mathbf{K} = \boldsymbol{\Omega}^{-1}, \quad \mathbf{a} = d\boldsymbol{\mu}, \quad \mathbf{H} = d\boldsymbol{\Omega} = \mathbf{H}'.$$

Its contribution to the criterion is

$$\ell_i(\boldsymbol{\mu}, \boldsymbol{\Omega}) = -\frac{m}{2} \log(2\pi) - \frac{1}{2} \log |\boldsymbol{\Omega}| - \frac{1}{2} \mathbf{z}' \mathbf{K} \mathbf{z}.$$

Its first differential is

$$d\ell_i = \mathbf{a}' \mathbf{K} \mathbf{z} + \frac{1}{2} \operatorname{tr}\{(\mathbf{K} \mathbf{z} \mathbf{z}' \mathbf{K} - \mathbf{K}) \mathbf{H}\}.$$

Using $d\mathbf{K} = -\mathbf{K}\mathbf{H}\mathbf{K}$ and $d\mathbf{z} = -\mathbf{a}$ gives

$$\begin{aligned} d^2\ell_i &= -\mathbf{a}'\mathbf{K}\mathbf{a} - 2\mathbf{a}'\mathbf{K}\mathbf{H}\mathbf{K}\mathbf{z} + \frac{1}{2}\text{tr}(\mathbf{K}\mathbf{H}\mathbf{K}\mathbf{H}) \\ &\quad - \text{tr}(\mathbf{K}\mathbf{z}\mathbf{z}'\mathbf{K}\mathbf{H}\mathbf{K}\mathbf{H}). \end{aligned}$$

Since $\mathbf{E}[\mathbf{z}] = \mathbf{0}$ and $\mathbf{E}[\mathbf{z}\mathbf{z}'] = \mathbf{\Omega} = \mathbf{K}^{-1}$,

$$-\mathbf{E}[d^2\ell_i] = \mathbf{a}'\mathbf{K}\mathbf{a} + \frac{1}{2}\text{tr}(\mathbf{K}\mathbf{H}\mathbf{K}\mathbf{H}).$$

Now

$$\text{vec } \mathbf{H} = \mathbf{D}_m d\boldsymbol{\omega}, \quad \text{tr}(\mathbf{K}\mathbf{H}\mathbf{K}\mathbf{H}) = (d\boldsymbol{\omega})'\mathbf{D}_m'(\mathbf{K} \otimes \mathbf{K})\mathbf{D}_m d\boldsymbol{\omega}.$$

Summing the n contributions and applying Theorem 14.3 gives the block matrix. □

3.3 OLS sensitivity and exact deletion

Theorem 14.16 (Differential of ordinary least squares). Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have full column rank, let $\mathbf{y} \in \mathbb{R}^n$, and put

$$\mathbf{Q} = \mathbf{X}'\mathbf{X}, \quad \mathbf{G} = \mathbf{Q}^{-1}\mathbf{X}', \quad \hat{\boldsymbol{\beta}} = \mathbf{G}\mathbf{y}, \quad \hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

For simultaneous increments $d\mathbf{X}$ and $d\mathbf{y}$,

$$d\hat{\boldsymbol{\beta}} = \mathbf{G} d\mathbf{y} + \mathbf{Q}^{-1}(d\mathbf{X})'\hat{\boldsymbol{\varepsilon}} - \mathbf{G}(d\mathbf{X})\hat{\boldsymbol{\beta}}.$$

Consequently,

$$D_{\text{vec } \mathbf{X}} \hat{\boldsymbol{\beta}} = \mathbf{Q}^{-1} \otimes \hat{\boldsymbol{\varepsilon}}' - \hat{\boldsymbol{\beta}}' \otimes \mathbf{G}, \quad D_{\mathbf{y}} \hat{\boldsymbol{\beta}} = \mathbf{G}.$$

Origin: Adapted from A&M Ex. 13.41

Proof. Differentiating $\widehat{\boldsymbol{\beta}} = \mathbf{Q}^{-1}\mathbf{X}'\mathbf{y}$ gives

$$\begin{aligned} d\widehat{\boldsymbol{\beta}} &= -\mathbf{Q}^{-1}\{(\mathbf{dX})'\mathbf{X} + \mathbf{X}'\mathbf{dX}\}\widehat{\boldsymbol{\beta}} + \mathbf{Q}^{-1}\{(\mathbf{dX})'\mathbf{y} + \mathbf{X}'\mathbf{dy}\} \\ &= \mathbf{G}\mathbf{dy} + \mathbf{Q}^{-1}(\mathbf{dX})'\widehat{\boldsymbol{\varepsilon}} - \mathbf{G}(\mathbf{dX})\widehat{\boldsymbol{\beta}}. \end{aligned}$$

The vec identities

$$(\mathbf{dX})'\widehat{\boldsymbol{\varepsilon}} = (\mathbf{I}_p \otimes \widehat{\boldsymbol{\varepsilon}}') \text{vec}(\mathbf{dX}), \quad \mathbf{G}(\mathbf{dX})\widehat{\boldsymbol{\beta}} = (\widehat{\boldsymbol{\beta}}' \otimes \mathbf{G}) \text{vec}(\mathbf{dX})$$

give the Jacobians. □

Theorem 14.17 (Exact row deletion). With the notation of Theorem 14.16, let $\tilde{\mathbf{x}}'_i$ be row i of $\mathbf{X}$, let

$$h_i = \tilde{\mathbf{x}}'_i \mathbf{Q}^{-1} \tilde{\mathbf{x}}_i, \quad \widehat{e}_i = y_i - \tilde{\mathbf{x}}'_i \widehat{\boldsymbol{\beta}},$$

let $\mathbf{X}_{-i}$ and $\mathbf{y}_{-i}$ delete observation i , and suppose $\text{rank } \mathbf{X}_{-i} = p$. Define

$$\widehat{\boldsymbol{\beta}}_{-i} = (\mathbf{X}'_{-i} \mathbf{X}_{-i})^{-1} \mathbf{X}'_{-i} \mathbf{y}_{-i}.$$

Then

$$0 \leq h_i < 1, \quad \hat{\boldsymbol{\beta}}_{-i} = \hat{\boldsymbol{\beta}} - \frac{\mathbf{Q}^{-1} \tilde{\mathbf{x}}_i \hat{e}_i}{1 - h_i}, \quad y_i - \tilde{\mathbf{x}}_i' \hat{\boldsymbol{\beta}}_{-i} = \frac{\hat{e}_i}{1 - h_i}.$$

Origin: Course formulation from H Cor. 18.2.10; CLM Ex. B1

Proof. Write $\mathbf{Q}_{-i} = \mathbf{Q} - \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i'$. This matrix is positive definite by the deletion assumption. If $\mathbf{v} = \mathbf{Q}^{-1} \tilde{\mathbf{x}}_i$, then

$$h_i = \mathbf{v}' \mathbf{Q} \mathbf{v} = \mathbf{v}' \mathbf{Q}_{-i} \mathbf{v} + h_i^2.$$

Thus $h_i = 0$ when $\tilde{\mathbf{x}}_i = \mathbf{0}$; otherwise $h_i(1 - h_i) > 0$. The rank-one inverse formula yields

$$\mathbf{Q}_{-i}^{-1} = \mathbf{Q}^{-1} + \frac{\mathbf{Q}^{-1} \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i' \mathbf{Q}^{-1}}{1 - h_i}.$$

Since $\mathbf{X}_{-i}' \mathbf{y}_{-i} = \mathbf{X}' \mathbf{y} - \tilde{\mathbf{x}}_i y_i$, substitution gives

$$\hat{\boldsymbol{\beta}}_{-i} - \hat{\boldsymbol{\beta}} = -\frac{\mathbf{Q}^{-1} \tilde{\mathbf{x}}_i (y_i - \tilde{\mathbf{x}}_i' \hat{\boldsymbol{\beta}})}{1 - h_i}.$$

Premultiplication by $\tilde{\mathbf{x}}_i'$ gives the held-out residual identity. □

4 Computation: Numerical Checks in R

4.1 Two Hessian checks (§1.2)

```
1 B <- ISLR2::Boston
2 Xh <- cbind(1, scale(as.matrix(B[, c("lstat", "rm")]])))
3 yh <- B$medv
4 qfun <- function(b) sum((yh - drop(Xh %*% b))^2)
5 b_at <- c(20, -0.4, 3)
6 H_manual <- 2 * crossprod(Xh)
7 H_package <- optimHess(
8   b_at, qfun,
9   control = list(ndeps = rep(0.1, length(b_at)))
10 )
11
12 Omega_h <- matrix(c(2, .3, .2,
13                     .3, 1.5, .1,
14                     .2, .1, 1.2), 3, 3, byrow = TRUE)
15 Delta_h <- matrix(c(1, .2, -.1,
16                     .2, -.4, .3,
17                     -.1, .3, .5), 3, 3, byrow = TRUE)
```

```

18 barrier <- function(A) -as.numeric(determinant(A, logarithm = TRUE)$modulus)
19 step_h <- 1e-4
20 second_fd <- (barrier(Omega_h + step_h * Delta_h) -
21             2 * barrier(Omega_h) +
22             barrier(Omega_h - step_h * Delta_h)) / step_h^2
23 K_h <- solve(Omega_h)
24 second_exact <- sum(diag(K_h %*% Delta_h %*% K_h %*% Delta_h))
25 tol_pd <- 1e-10
26 eig_barrier <- eigen(Omega_h, symmetric = TRUE, only.values = TRUE)$values
27 c(min_eigenvalue_barrier_point = min(eig_barrier),
28   positive_definite_tolerance = tol_pd,
29   least_squares_max_abs = max(abs(H_manual - H_package)),
30   logdet_exact = second_exact,
31   logdet_finite_difference = second_fd,
32   logdet_abs_difference = abs(second_exact - second_fd))

```

```

## min_eigenvalue_barrier_point  positive_definite_tolerance
##                1.149e+00                1.000e-10
##      least_squares_max_abs                logdet_exact
##                1.273e-10                6.849e-01
##      logdet_finite_difference                logdet_abs_difference

```

##

6.849e-01

1.319e-08

4.2 OLS normal equations against 1m QR (§2.1)

```

1 X <- model.matrix(~ lstat + rm + age, data = B)
2 y <- B$medv
3 fit <- lm(medv ~ lstat + rm + age, data = B)
4 b_normal <- drop(solve(crossprod(X), crossprod(X, y)))
5 tol_rank <- 1e-10
6 b_qr <- drop(qr.coef(qr(X, tol = tol_rank), y))
7 e_normal <- y - drop(X %*% b_normal)
8 c(qr_rank_at_tolerance = qr(X, tol = tol_rank)$rank,
9   max_normal_minus_lm = max(abs(b_normal - coef(fit))),
10  max_qr_minus_lm = max(abs(b_qr - coef(fit))),
11  normal_equation_residual = max(abs(crossprod(X, e_normal))))

```

```

##      qr_rank_at_tolerance      max_normal_minus_lm
##              4.000e+00              1.335e-12
##      max_qr_minus_lm normal_equation_residual
##              0.000e+00              3.790e-10

```

4.3 GLS against a whitened QR fit (§2.2)

```
1 rows_g <- seq_len(80)
2 Xg <- model.matrix(~ lstat + rm, data = B[rows_g, ])
3 yg <- B$medv[rows_g]
4 n_g <- nrow(Xg)
5 rho_g <- 0.6
6 Omega_g <- rho_g^abs(outer(seq_len(n_g), seq_len(n_g), "-"))
7 OiX <- solve(Omega_g, Xg)
8 Oiy <- solve(Omega_g, yg)
9 Q_g <- crossprod(Xg, OiX)
10 V_g <- solve(Q_g)
11 b_gls <- drop(V_g %*% crossprod(Xg, Oiy))
12
13 R_g <- chol(Omega_g)
14 X_white <- forwardsolve(t(R_g), Xg)
15 y_white <- drop(forwardsolve(t(R_g), yg))
16 b_white <- drop(lm.fit(X_white, y_white)$coefficients)
17 e_gls <- yg - drop(Xg %*% b_gls)
18 eig_Omega <- eigen(Omega_g, symmetric = TRUE, only.values = TRUE)$values
19 c(min_eigenvalue_Omega = min(eig_Omega),
```

```

20 positive_definite_tolerance = tol_rank,
21 whitened_qr_rank_at_tolerance = qr(X_white, tol = tol_rank)$rank,
22 max_gls_minus_whitened = max(abs(b_gls - b_white)),
23 weighted_normal_residual =
24   max(abs(crossprod(Xg, solve(Omega_g, e_gls))))

```

```

##           min_eigenvalue_Omega   positive_definite_tolerance
##                    2.501e-01                    1.000e-10
## whitened_qr_rank_at_tolerance      max_gls_minus_whitened
##                    3.000e+00                    1.908e-12
##           weighted_normal_residual
##                    6.037e-11

```

4.4 Restricted GLS against a block solve (§2.3)

```

1 R_con <- matrix(c(0, 1, 1), nrow = 1)
2 c_con <- 0
3 S_con <- drop(R_con %*% V_g %*% t(R_con))
4 g_con <- drop(R_con %*% b_gls - c_con)
5 b_restricted <- drop(b_gls -
6   V_g %*% t(R_con) %*% (g_con / S_con))
7

```

```

8 s_g <- drop(crossprod(Xg, 0iy))
9 KKT <- rbind(cbind(Q_g, t(R_con)),
10             cbind(R_con, matrix(0, 1, 1)))
11 b_block <- drop(solve(KKT, c(s_g, c_con))[seq_len(ncol(Xg))])
12 q_gls <- function(b) {
13   e <- yg - drop(Xg %*% b)
14   drop(crossprod(e, solve(Omega_g, e)))
15 }
16 increase_formula <- g_con^2 / S_con
17 reduction <- V_g %*% t(R_con) %*% (1 / S_con) %*% R_con %*% V_g
18 c(max_formula_minus_block = max(abs(b_restricted - b_block)),
19   constraint_residual = max(abs(R_con %*% b_restricted - c_con)),
20   objective_identity_residual =
21     abs(q_gls(b_restricted) - q_gls(b_gls) - increase_formula),
22   min_eigen_covariance_reduction =
23     min(eigen((reduction + t(reduction)) / 2,
24               symmetric = TRUE, only.values = TRUE)$values),
25   eigenvalue_tolerance = tol_rank)

```

##	max_formula_minus_block	constraint_residual
##	2.309e-13	5.551e-16

```
##      objective_identity_residual min_eigen_covariance_reduction
##              1.558e-11                      -2.818e-18
##      eigenvalue_tolerance
##              1.000e-10
```

4.5 Mean, covariance, and information (§3.1–§3.2)

```
1 Y <- unclass(scale(as.matrix(B[, c("medv", "crim", "lstat")])))
2 n_y <- nrow(Y)
3 m_y <- ncol(Y)
4 mu_manual <- drop(crossprod(rep(1 / n_y, n_y), Y))
5 Z_y <- sweep(Y, 2, mu_manual)
6 Omega_manual <- crossprod(Z_y) / n_y
7 Omega_package <- cov(Y) * (n_y - 1) / n_y
8
9 lower_y <- lower.tri(matrix(0, m_y, m_y), diag = TRUE)
10 vech_to_sym <- function(w) {
11   A <- matrix(0, m_y, m_y)
12   A[lower_y] <- w
13   A + t(A) - diag(diag(A))
14 }
```

```
15 D_y <- matrix(0, m_y^2, sum(lower_y))
16 for (j in seq_len(ncol(D_y))) {
17   E <- matrix(0, m_y, m_y)
18   E[which(lower_y)[j]] <- 1
19   E <- E + t(E) - diag(diag(E))
20   D_y[, j] <- as.vector(E)
21 }
22 nll <- function(theta) {
23   mu <- theta[seq_len(m_y)]
24   Omega <- vech_to_sym(theta[-seq_len(m_y)])
25   Z <- sweep(Y, 2, mu)
26   K <- solve(Omega)
27   n_y / 2 * as.numeric(determinant(Omega, logarithm = TRUE)$modulus) +
28   0.5 * sum((Z %*% K) * Z)
29 }
30 theta_hat <- c(mu_manual, Omega_manual[lower_y])
31 H_package_criterion <- optimHess(
32   theta_hat, nll,
33   control = list(ndeps = rep(3e-5, length(theta_hat)))
34 )
35 K_y <- solve(Omega_manual)
```

```

36 I_manual <- matrix(0, length(theta_hat), length(theta_hat))
37 I_manual[seq_len(m_y), seq_len(m_y)] <- n_y * K_y
38 omega_idx <- m_y + seq_len(ncol(D_y))
39 I_manual[omega_idx, omega_idx] <-
40   n_y / 2 * t(D_y) %*% kronecker(K_y, K_y) %*% D_y
41 score_Omega <- 0.5 *
42   (K_y %*% crossprod(Z_y) %*% K_y - n_y * K_y)
43 eig_Omega_criterion <- eigen(Omega_manual, symmetric = TRUE,
44                             only.values = TRUE)$values
45 c(min_eigenvalue_covariance = min(eig_Omega_criterion),
46   positive_definite_tolerance = tol_rank,
47   max_mean_minus_colMeans = max(abs(mu_manual - colMeans(Y))),
48   max_covariance_minus_cov = max(abs(Omega_manual - Omega_package)),
49   covariance_score_residual = max(abs(score_Omega)),
50   information_minus_optimHess = max(abs(I_manual - H_package_criterion)),
51   information_relative_residual =
52     max(abs(I_manual - H_package_criterion)) / max(abs(I_manual)))

```

```

##      min_eigenvalue_covariance      positive_definite_tolerance
##                2.573e-01                1.000e-10
##      max_mean_minus_colMeans      max_covariance_minus_cov

```

```
##                9.103e-17                3.331e-16
## covariance_score_residual    information_minus_optimHess
##                2.842e-13                2.958e-04
## information_relative_residual
##                7.578e-08
```

4.6 Differential and exact row deletion (§3.3)

```
1 Xs <- cbind(1, scale(as.matrix(B[, c("lstat", "rm", "age")])))
2 ys <- B$medv
3 Q_s <- crossprod(Xs)
4 G_s <- solve(Q_s, t(Xs))
5 b_s <- drop(G_s %*% ys)
6 e_s <- ys - drop(Xs %*% b_s)
7 DX <- matrix(sin(seq_along(Xs)) / 10, nrow(Xs), ncol(Xs))
8 dy <- cos(seq_along(ys)) / 10
9 db_formula <- drop(G_s %*% dy +
10   solve(Q_s, crossprod(DX, e_s)) - G_s %*% DX %*% b_s)
11 step_s <- 1e-6
12 b_perturbed <- drop(lm.fit(Xs + step_s * DX,
13   ys + step_s * dy)$coefficients)
```

```

14 db_finite <- (b_perturbed - b_s) / step_s
15
16 leverages <- rowSums((Xs %*% solve(Q_s)) * Xs)
17 i_del <- which.max(leverages)
18 x_i <- Xs[i_del, ]
19 h_i <- leverages[i_del]
20 b_delete_formula <- drop(b_s -
21   solve(Q_s, x_i) * e_s[i_del] / (1 - h_i))
22 b_delete_package <- drop(lm.fit(Xs[-i_del, ], ys[-i_del])$coefficients)
23 heldout_package <- ys[i_del] - drop(x_i %*% b_delete_package)
24 c(finite_difference_residual = max(abs(db_finite - db_formula)),
25   deletion_coefficient_residual =
26     max(abs(b_delete_formula - b_delete_package)),
27   heldout_residual_identity =
28     abs(heldout_package - e_s[i_del] / (1 - h_i)),
29   deletion_denominator = 1 - h_i)

```

```

##      finite_difference_residual deletion_coefficient_residual
##                1.456e-07                7.105e-15
## heldout_residual_identity.366      deletion_denominator.366
##                5.329e-15                9.229e-01

```

5 Pitfalls

5.1 Reading a nonsymmetric coefficient as a Hessian

$$d^2f = (d\mathbf{x})'\mathbf{B} d\mathbf{x}, \quad \mathbf{H}f = \frac{\mathbf{B} + \mathbf{B}'}{2}.$$

The second equality is Theorem 14.3; $\mathbf{B}$ itself is a Hessian only when it is symmetric.

5.2 Dropping the curvature term in a composition

For $f(u) = u$ and $g(x) = x^2$,

$$g'(x)^2 f''(g(x)) = 0, \quad (f \circ g)''(x) = 2.$$

The missing term in Theorem 14.5 is $f'(g(x))g''(x)$.

5.3 Classifying from a zero Hessian

For $f(x) = x^3$,

$$f'(0) = f''(0) = 0, \quad f(-t) < f(0) < f(t) \quad (t > 0).$$

A semidefinite Hessian at one stationary point is inconclusive.

5.4 Replacing a global comparison by a local test

Negative definiteness of $\mathbf{H}\ell$ at a stationary point proves only a strict local maximum. The eigenvalue comparison in Theorem 14.13 proves the global claim.

5.5 Differentiating duplicate covariance entries independently

For $\mathbf{\Omega} = \mathbf{\Omega}'$, Theorem 14.4 gives

$$\text{vec}(\text{d}\mathbf{\Omega}) = \mathbf{D}_m \text{d}(\text{vech } \mathbf{\Omega}), \quad \mathbf{H}_{\text{vech } \mathbf{\Omega}} f = \mathbf{D}_m' \mathbf{H}_{\text{vec } \mathbf{\Omega}} f \mathbf{D}_m.$$

5.6 Using an identity-scaled formula under general moments

With $\mathbf{G}_0$ from Theorem 14.9, if $\text{cov}(\boldsymbol{\varepsilon}) = \mathbf{\Omega}$, then

$$\text{cov}(\hat{\boldsymbol{\beta}}_{\text{OLS}}) = \mathbf{G}_0 \mathbf{\Omega} \mathbf{G}_0',$$

not $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ unless the stated covariance justifies it.

5.7 Deleting a row when rank is lost

$$\mathbf{Q}_{-i} := \mathbf{X}_{-i}' \mathbf{X}_{-i} = \mathbf{Q} - \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i', \quad \mathbf{v} := \mathbf{Q}^{-1} \tilde{\mathbf{x}}_i.$$

With the notation of Theorem 14.17, $h_i = 1$ implies

$$\mathbf{v} \neq \mathbf{0}, \quad \mathbf{v}'\mathbf{Q}_{-i}\mathbf{v} = \|\mathbf{X}_{-i}\mathbf{v}\|_2^2 = h_i(1 - h_i) = 0, \quad \mathbf{Q}_{-i}\mathbf{v} = \mathbf{X}_{-i}'\mathbf{X}_{-i}\mathbf{v} = \mathbf{0}.$$

Thus $\text{rank } \mathbf{X}_{-i} < p$ and the leave-one-out coefficient vector is not uniquely defined by least squares.

5.8 Turning a criterion into unstated distribution theory

Definition 14.12 specifies an objective function. Theorem 14.15 uses only $\mathbf{E}[\mathbf{y}_i] = \boldsymbol{\mu}$ and $\text{cov}(\mathbf{y}_i) = \boldsymbol{\Omega}$; it supplies no sampling distribution.

6 Exercises

Exercise 14.1 (Affine pullback of a Hessian). Let $f : \mathcal{C} \rightarrow \mathbb{R}$ have continuous second partial derivatives on an open convex set $\mathcal{C} \subseteq \mathbb{R}^q$. Let $\mathbf{A} \in \mathbb{R}^{q \times p}$, $\mathbf{c} \in \mathbb{R}^q$, and

$$g(\mathbf{x}) = f(\mathbf{Ax} + \mathbf{c})$$

on $\mathcal{U} = \{\mathbf{x} : \mathbf{Ax} + \mathbf{c} \in \mathcal{C}\}$. Prove that $\mathcal{U}$ is open and convex, and prove

$$\nabla g(\mathbf{x}) = \mathbf{A}' \nabla f(\mathbf{Ax} + \mathbf{c}), \quad \mathbf{H}g(\mathbf{x}) = \mathbf{A}' \mathbf{H}f(\mathbf{Ax} + \mathbf{c}) \mathbf{A}.$$

Deduce that f convex implies g convex. If every Hessian of f is positive definite and $\text{rank } \mathbf{A} = p$, prove that g is strictly convex. Construct a strictly convex f and a rank-deficient $\mathbf{A}$ for which g is not strictly convex.

Origin: Course formulation from MC Prop. 14

Solution. For $\mathbf{x}, \mathbf{z} \in \mathcal{U}$ and $0 \leq t \leq 1$,

$$\mathbf{A}\{(1-t)\mathbf{x} + t\mathbf{z}\} + \mathbf{c} = (1-t)(\mathbf{Ax} + \mathbf{c}) + t(\mathbf{Az} + \mathbf{c}) \in \mathcal{C},$$

so $\mathcal{U}$ is convex. Fix $\mathbf{x} \in \mathcal{U}$. Since $\mathcal{C}$ is open, some $\rho > 0$ satisfies

$$\|\mathbf{y} - (\mathbf{Ax} + \mathbf{c})\|_2 < \rho \implies \mathbf{y} \in \mathcal{C}.$$

If $\|\mathbf{h}\|_2 < \rho/(1 + \|\mathbf{A}\|_2)$, then

$$\|\mathbf{Ah}\|_2 \leq \|\mathbf{A}\|_2 \|\mathbf{h}\|_2 < \rho,$$

so $\mathbf{x} + \mathbf{h} \in \mathcal{U}$. Hence $\mathcal{U}$ is open.

For $\mathbf{y} = \mathbf{Ax} + \mathbf{c}$,

$$d\mathbf{y} = \mathbf{A} d\mathbf{x}, \quad d^2 y_j = 0 \quad (j = 1, \dots, q).$$

The earlier first-order chain rule and Theorem 14.5 give the two identities. If f is convex, then $\mathbf{H}f(\mathbf{y}) \succeq \mathbf{O}$ for every $\mathbf{y} \in \mathcal{C}$, so

$$\mathbf{h}'\mathbf{H}g(\mathbf{x})\mathbf{h} = (\mathbf{Ah})'\mathbf{H}f(\mathbf{Ax} + \mathbf{c})(\mathbf{Ah}) \geq 0.$$

If every $\mathbf{H}f(\mathbf{y}) \succ \mathbf{O}$ and $\text{rank } \mathbf{A} = p$, then

$$\mathbf{h} \neq \mathbf{0} \implies \mathbf{Ah} \neq \mathbf{0} \implies \mathbf{h}'\mathbf{H}g\mathbf{h} > 0.$$

Apply Theorem 14.7. For $q = 1$, $p = 2$, take

$$\mathcal{C} = \mathbb{R}, \quad f(y) = y^2, \quad \mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad \mathbf{c} = 0.$$

Then $f''(y) = 2 > 0$, whereas

$$g(x_1, x_2) = x_1^2, \quad g(x_1, x_2 + t) = g(x_1, x_2) \quad (t \in \mathbb{R}),$$

so g is not strictly convex. 

Exercise 14.2 (Exact nonlinear least-squares Hessian). Let $\mathcal{U} \subseteq \mathbb{R}^p$ be open, and let $\mathbf{r} : \mathcal{U} \rightarrow \mathbb{R}^n$ have continuous second partial derivatives, let $\mathbf{W} = \mathbf{W}' \in \mathbb{R}^{n \times n}$ with $\mathbf{W} \succeq \mathbf{O}$, and define

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{r}(\mathbf{x})' \mathbf{W} \mathbf{r}(\mathbf{x}), \quad \mathbf{J}(\mathbf{x}) = \text{Dr}(\mathbf{x}).$$

Prove

$$\nabla f(\mathbf{x}) = \mathbf{J}(\mathbf{x})' \mathbf{W} \mathbf{r}(\mathbf{x}),$$

$$\mathbf{H}f(\mathbf{x}) = \mathbf{J}(\mathbf{x})' \mathbf{W} \mathbf{J}(\mathbf{x}) + \sum_{i=1}^n \{\mathbf{W} \mathbf{r}(\mathbf{x})\}_i \mathbf{H}r_i(\mathbf{x}).$$

Prove that every $\mathbf{x}_*$ satisfying $\mathbf{r}(\mathbf{x}_*) = \mathbf{0}$ is a global minimiser. If also $\mathbf{W} \succ \mathbf{0}$ and $\text{rank } \mathbf{J}(\mathbf{x}_*) = p$, prove that the minimum is locally strict.

Origin: Adapted from A&M Ex. 13.53; MC §10

Solution. For $q(\mathbf{u}) = \frac{1}{2} \mathbf{u}' \mathbf{W} \mathbf{u}$,

$$\nabla q(\mathbf{u}) = \mathbf{W} \mathbf{u}, \quad \mathbf{H}q(\mathbf{u}) = \mathbf{W}.$$

The earlier first-order chain rule and Theorem 14.5 give the displayed formulas. If $\mathbf{r}(\mathbf{x}_*) = \mathbf{0}$, then

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{r}(\mathbf{x})' \mathbf{W} \mathbf{r}(\mathbf{x}) \geq 0 = f(\mathbf{x}_*),$$

so $\mathbf{x}_*$ is a global minimiser. Under the additional hypotheses,

$$\nabla f(\mathbf{x}_*) = \mathbf{0}, \quad \mathbf{H}f(\mathbf{x}_*) = \mathbf{J}(\mathbf{x}_*)' \mathbf{W} \mathbf{J}(\mathbf{x}_*) \succ \mathbf{0},$$

because $\mathbf{J}(\mathbf{x}_*)\mathbf{h} \neq \mathbf{0}$ for $\mathbf{h} \neq \mathbf{0}$. Thus the global minimum is locally strict by Theorem 14.7(d). ■

Exercise 14.3 (When a semidefinite quadratic is bounded below). Let $\mathbf{V} = \mathbf{V}' \in \mathbb{R}^{p \times p}$ with $\mathbf{V} \succeq \mathbf{0}$, let $\mathbf{b} \in \mathbb{R}^p$, and define

$$q(\mathbf{a}) = \mathbf{a}' \mathbf{V} \mathbf{a} - 2\mathbf{b}' \mathbf{a}.$$

Prove that q is bounded below if and only if $\mathbf{b} \in \text{col } \mathbf{V}$. In that case,

$$\arg \min_{\mathbf{a}} q(\mathbf{a}) = \{\mathbf{a} : \mathbf{V} \mathbf{a} = \mathbf{b}\}, \quad \min_{\mathbf{a}} q(\mathbf{a}) = -\mathbf{b}' \mathbf{a}_*$$

for any $\mathbf{a}_*$ satisfying $\mathbf{V} \mathbf{a}_* = \mathbf{b}$.

Origin: Adapted from H Ex. 19.1 and Thm. 19.1.1

Solution. If $\mathbf{b} = \mathbf{V}\mathbf{a}_*$, then

$$q(\mathbf{a}) - q(\mathbf{a}_*) = (\mathbf{a} - \mathbf{a}_*)'\mathbf{V}(\mathbf{a} - \mathbf{a}_*) \geq 0.$$

Equality holds exactly when $\mathbf{V}(\mathbf{a} - \mathbf{a}_*) = \mathbf{0}$, equivalently $\mathbf{V}\mathbf{a} = \mathbf{b}$, and

$$q(\mathbf{a}_*) = \mathbf{a}_*'\mathbf{b} - 2\mathbf{b}'\mathbf{a}_* = -\mathbf{b}'\mathbf{a}_*.$$

If $\mathbf{b} \notin \text{col } \mathbf{V} = (\text{null } \mathbf{V})^\perp$, choose $\mathbf{z} \in \text{null } \mathbf{V}$ with $\mathbf{b}'\mathbf{z} \neq 0$ and replace $\mathbf{z}$ by $-\mathbf{z}$ if necessary so that $\mathbf{b}'\mathbf{z} > 0$. For $t > 0$,

$$q(t\mathbf{z}) = -2t\mathbf{b}'\mathbf{z} \longrightarrow -\infty.$$



Exercise 14.4 (Trace minus log-determinant). Let $\mathbf{A} = \mathbf{A}' \succ \mathbf{O}$ be $m \times m$ and, for $\mathbf{\Omega} = \mathbf{\Omega}' \succ \mathbf{O}$, define

$$\phi(\mathbf{\Omega}) = \text{tr}(\mathbf{A}\mathbf{\Omega}) - \log |\mathbf{\Omega}|.$$

Prove that ϕ is strictly convex and has the unique global minimiser

$$\mathbf{\Omega}_* = \mathbf{A}^{-1}, \quad \phi(\mathbf{\Omega}_*) = m + \log |\mathbf{A}|.$$

Origin: Course formulation from A&M Ex. 13.52; H Eq. (15.9.3)

Solution. Example 14.2 proves that the positive-definite cone is open and convex. The linear term has zero second differential. The earlier inverse- and log-determinant differential rules, with $\mathbf{\Omega} \succ \mathbf{O}$, give

$$d\phi = \text{tr}\{(\mathbf{A} - \mathbf{\Omega}^{-1})d\mathbf{\Omega}\}, \quad d^2\phi = \text{tr}\{\mathbf{\Omega}^{-1}(d\mathbf{\Omega})\mathbf{\Omega}^{-1}(d\mathbf{\Omega})\} > 0$$

for every nonzero symmetric increment. Thus ϕ is strictly convex. At $\mathbf{\Omega}_* = \mathbf{A}^{-1}$, $d\phi = 0$ for every symmetric increment, so Theorem 14.7 gives the unique global minimum. Its value is

$$\text{tr}(\mathbf{I}_m) - \log |\mathbf{A}^{-1}| = m + \log |\mathbf{A}|. \quad \blacksquare$$

Exercise 14.5 (An affine log-determinant barrier). Let $\mathbf{A} = \mathbf{A}'$ and $\mathbf{B}_1, \dots, \mathbf{B}_p$ be symmetric $m \times m$ matrices. On

$$\mathcal{U} = \left\{ \mathbf{x} \in \mathbb{R}^p : \mathbf{F}(\mathbf{x}) = \mathbf{A} + \sum_{j=1}^p x_j \mathbf{B}_j \succ \mathbf{0} \right\},$$

assume $\mathcal{U} \neq \emptyset$, and put $f(\mathbf{x}) = -\log |\mathbf{F}(\mathbf{x})|$. Prove

$$\begin{aligned} \frac{\partial f}{\partial x_i} &= -\operatorname{tr}\{\mathbf{F}(\mathbf{x})^{-1} \mathbf{B}_i\}, \\ \frac{\partial^2 f}{\partial x_i \partial x_j} &= \operatorname{tr}\{\mathbf{F}(\mathbf{x})^{-1} \mathbf{B}_i \mathbf{F}(\mathbf{x})^{-1} \mathbf{B}_j\}. \end{aligned}$$

Prove that the Hessian is positive semidefinite, and is positive definite if and only if $\mathbf{B}_1, \dots, \mathbf{B}_p$ are linearly independent.

Origin: Course formulation from H Eq. (15.9.3); A&M Ex. 13.52

Solution. The positive-definite cone is open by Example 14.2; the affine-preimage argument in Exercise 14.1 makes $\mathcal{U}$ open. The earlier inverse- and log-determinant differential rules apply because $\mathbf{F}(\mathbf{x}) \succ \mathbf{O}$. Moreover,

$$\frac{\partial \mathbf{F}}{\partial x_i} = \mathbf{B}_i, \quad \frac{\partial \mathbf{F}^{-1}}{\partial x_j} = -\mathbf{F}^{-1} \mathbf{B}_j \mathbf{F}^{-1},$$

$$\begin{aligned} \frac{\partial f}{\partial x_i} &= -\operatorname{tr}(\mathbf{F}^{-1} \mathbf{B}_i), \\ \frac{\partial^2 f}{\partial x_i \partial x_j} &= -\operatorname{tr} \left\{ \frac{\partial \mathbf{F}^{-1}}{\partial x_j} \mathbf{B}_i \right\} = \operatorname{tr}(\mathbf{F}^{-1} \mathbf{B}_j \mathbf{F}^{-1} \mathbf{B}_i) \\ &= \operatorname{tr}(\mathbf{F}^{-1} \mathbf{B}_i \mathbf{F}^{-1} \mathbf{B}_j). \end{aligned}$$

For $\mathbf{h} \in \mathbb{R}^p$, set $\mathbf{H} = \sum_i h_i \mathbf{B}_i$. Then

$$\mathbf{h}' \mathbf{H} f(\mathbf{x}) \mathbf{h} = \operatorname{tr}(\mathbf{F}^{-1} \mathbf{H} \mathbf{F}^{-1} \mathbf{H}) = \operatorname{tr}\{(\mathbf{F}^{-1/2} \mathbf{H} \mathbf{F}^{-1/2})^2\} \geq 0.$$

Equality holds exactly when $\mathbf{H} = \mathbf{O}$. Thus the Hessian is positive definite exactly when

$$\sum_i h_i \mathbf{B}_i = \mathbf{O} \implies \mathbf{h} = \mathbf{0},$$

which is linear independence of the $\mathbf{B}_i$. ■

Exercise 14.6 (Equality-constrained sensitivity). Let $\mathbf{Q} = \mathbf{Q}' \succ \mathbf{O}$, let $1 \leq q \leq p$, and let $\mathbf{R} \in \mathbb{R}^{q \times p}$ have rank q . Let $\mathbf{s} \in \mathbb{R}^p$. For each $\mathbf{c} \in \mathbb{R}^q$, minimise

$$f(\boldsymbol{\beta}) = \frac{1}{2} \boldsymbol{\beta}' \mathbf{Q} \boldsymbol{\beta} - \mathbf{s}' \boldsymbol{\beta} \quad \text{subject to} \quad \mathbf{R} \boldsymbol{\beta} = \mathbf{c}.$$

Set $\mathbf{S} = \mathbf{R} \mathbf{Q}^{-1} \mathbf{R}'$. Find $\boldsymbol{\beta}(\mathbf{c})$ and $\boldsymbol{\lambda}(\mathbf{c})$ satisfying

$$\mathbf{R} \boldsymbol{\beta}(\mathbf{c}) = \mathbf{c}, \quad \mathbf{Q} \boldsymbol{\beta}(\mathbf{c}) - \mathbf{s} + \mathbf{R}' \boldsymbol{\lambda}(\mathbf{c}) = \mathbf{0}.$$

Prove that $\boldsymbol{\beta}(\mathbf{c})$ is the minimiser, find the minimum value $v(\mathbf{c})$, and show

$$\nabla_{\mathbf{c}} v(\mathbf{c}) = -\boldsymbol{\lambda}(\mathbf{c}), \quad \mathbf{H}_{\mathbf{c}} v(\mathbf{c}) = \mathbf{S}^{-1} \succ \mathbf{O}.$$

Origin: Course formulation from A&M Ex. 13.56; H Thm. 19.2.1

Solution. Put $\beta_0 = \mathbf{Q}^{-1}\mathbf{s}$. For $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{a}'\mathbf{S}\mathbf{a} = (\mathbf{R}'\mathbf{a})'\mathbf{Q}^{-1}(\mathbf{R}'\mathbf{a}) > 0,$$

so $\mathbf{S} \succ \mathbf{0}$. The equations

$$\mathbf{Q}\beta - \mathbf{s} + \mathbf{R}'\lambda = \mathbf{0}, \quad \mathbf{R}\beta = \mathbf{c}$$

have the unique solution

$$\begin{aligned} \lambda(\mathbf{c}) &= \mathbf{S}^{-1}(\mathbf{R}\beta_0 - \mathbf{c}), \\ \beta(\mathbf{c}) &= \beta_0 - \mathbf{Q}^{-1}\mathbf{R}'\mathbf{S}^{-1}(\mathbf{R}\beta_0 - \mathbf{c}). \end{aligned}$$

For every feasible $\beta = \beta(\mathbf{c}) + \mathbf{d}$, $\mathbf{R}\mathbf{d} = \mathbf{0}$ and

$$f(\beta) - f(\beta(\mathbf{c})) = \mathbf{d}'\{\mathbf{Q}\beta(\mathbf{c}) - \mathbf{s}\} + \frac{1}{2}\mathbf{d}'\mathbf{Q}\mathbf{d} = -\lambda'\mathbf{R}\mathbf{d} + \frac{1}{2}\mathbf{d}'\mathbf{Q}\mathbf{d} = \frac{1}{2}\mathbf{d}'\mathbf{Q}\mathbf{d},$$

so this solution is the unique minimiser. Completing the square gives

$$v(\mathbf{c}) = -\frac{1}{2}\mathbf{s}'\mathbf{Q}^{-1}\mathbf{s} + \frac{1}{2}(\mathbf{R}\beta_0 - \mathbf{c})'\mathbf{S}^{-1}(\mathbf{R}\beta_0 - \mathbf{c}).$$

Differentiation gives

$$\nabla_{\mathbf{c}} v = \mathbf{S}^{-1}(\mathbf{c} - \mathbf{R}\beta_0) = -\boldsymbol{\lambda}(\mathbf{c}), \quad \mathbf{H}_{\mathbf{c}} v = \mathbf{S}^{-1}.$$



Exercise 14.7 (When OLS and GLS fitted values agree). Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have full column rank and let $\boldsymbol{\Omega} = \boldsymbol{\Omega}' \in \mathbb{R}^{n \times n}$ with $\boldsymbol{\Omega} \succ \mathbf{O}$. Define

$$\mathbf{P}_0 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad \mathbf{P}_{\boldsymbol{\Omega}} = \mathbf{X}(\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}.$$

Prove

$$\mathbf{P}_0 = \mathbf{P}_{\boldsymbol{\Omega}} \iff \boldsymbol{\Omega}^{-1}\mathbf{X} = \mathbf{X}\mathbf{A} \quad \text{for some } \mathbf{A} \in \mathbb{R}^{p \times p}.$$

Origin: Adapted from H Thm. 14.12.18

Solution. If $\mathbf{P}_0 = \mathbf{P}_{\boldsymbol{\Omega}}$, Theorem 14.10 gives

$$\mathbf{X}'\boldsymbol{\Omega}^{-1}(\mathbf{I} - \mathbf{P}_0) = \mathbf{O}.$$

After transposition,

$$(\mathbf{I} - \mathbf{P}_0)\boldsymbol{\Omega}^{-1}\mathbf{X} = \mathbf{O},$$

so every column of $\boldsymbol{\Omega}^{-1}\mathbf{X}$ belongs to $\text{col } \mathbf{X}$; hence $\boldsymbol{\Omega}^{-1}\mathbf{X} = \mathbf{X}\mathbf{A}$ for some $\mathbf{A}$.

Conversely, this identity and symmetry give

$$\mathbf{X}'\boldsymbol{\Omega}^{-1}(\mathbf{I} - \mathbf{P}_0) = \mathbf{A}'\mathbf{X}'(\mathbf{I} - \mathbf{P}_0) = \mathbf{O}.$$

Thus $\mathbf{P}_0\mathbf{y} \in \text{col } \mathbf{X}$ satisfies the weighted normal equation for every $\mathbf{y}$. Uniqueness in Theorem 14.10 gives $\mathbf{P}_0\mathbf{y} = \mathbf{P}_\Omega\mathbf{y}$ for every $\mathbf{y}$, hence equality of the matrices. 

Exercise 14.8 (A BLUE linear function under rank deficiency). Let

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}, \quad \text{cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n,$$

where $\mathbf{X} \in \mathbb{R}^{n \times p}$ may be rank deficient, $\boldsymbol{\beta} \in \mathbb{R}^p$, and $\sigma^2 > 0$. Let $\mathbf{a} \in \text{col } \mathbf{X}'$ and set

$$\mathbf{g}_0 = \mathbf{X}(\mathbf{X}'\mathbf{X})^+\mathbf{a}.$$

Prove that $\mathbf{g}'_0\mathbf{y}$ is the unique minimum-variance linear unbiased estimator of $\mathbf{a}'\boldsymbol{\beta}$.

Origin: Course formulation from A&M Ex. 13.59; H Thm. 20.3.6

Solution. The earlier Moore–Penrose range identities give

$$\text{col}(\mathbf{X}'\mathbf{X}) = \text{col } \mathbf{X}', \quad \mathbf{X}'\mathbf{g}_0 = \mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^+\mathbf{a} = \mathbf{a}.$$

Thus $\mathbf{E}[\mathbf{g}'_0\mathbf{y}] = \mathbf{a}'\boldsymbol{\beta}$. Any $\mathbf{g}'\mathbf{y}$ is unbiased for $\mathbf{a}'\boldsymbol{\beta}$ exactly when $\mathbf{X}'\mathbf{g} = \mathbf{a}$. For $\mathbf{d} = \mathbf{g} - \mathbf{g}_0$, this condition gives $\mathbf{X}'\mathbf{d} = \mathbf{0}$, and hence

$$\mathbf{g}'_0\mathbf{d} = \mathbf{a}'(\mathbf{X}'\mathbf{X})^+\mathbf{X}'\mathbf{d} = 0.$$

Therefore

$$\text{var}(\mathbf{g}'\mathbf{y}) - \text{var}(\mathbf{g}'_0\mathbf{y}) = \sigma^2(\|\mathbf{g}\|_2^2 - \|\mathbf{g}_0\|_2^2) = \sigma^2\|\mathbf{d}\|_2^2 \geq 0,$$

with equality only at $\mathbf{g} = \mathbf{g}_0$. 

Exercise 14.9 (Sensitivity of GLS to its covariance matrix). Fix $\mathbf{X} \in \mathbb{R}^{n \times p}$ of full column rank and $\mathbf{y} \in \mathbb{R}^n$. Let $\mathbf{\Omega} = \mathbf{\Omega}' \in \mathbb{R}^{n \times n}$ vary within the positive-definite cone. With

$$\mathbf{V} = (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}, \quad \hat{\boldsymbol{\beta}}_G = \mathbf{V}\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y}, \quad \hat{\boldsymbol{\varepsilon}}_G = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_G,$$

prove, for symmetric increments $d\mathbf{\Omega} = d\mathbf{\Omega}'$,

$$d\hat{\boldsymbol{\beta}}_G = -\mathbf{V}\mathbf{X}'\mathbf{\Omega}^{-1}(d\mathbf{\Omega})\mathbf{\Omega}^{-1}\hat{\boldsymbol{\varepsilon}}_G.$$

Deduce that a scalar rescaling $d\mathbf{\Omega} = c\mathbf{\Omega}$ has zero first-order effect on $\hat{\boldsymbol{\beta}}_G$.

Origin: Adapted from A&M Ex. 13.43

Solution. Theorem 14.10 gives the weighted normal equation

$$\mathbf{X}'\mathbf{\Omega}^{-1}\hat{\boldsymbol{\varepsilon}}_G = \mathbf{0}.$$

Since $\mathbf{\Omega} \succ \mathbf{0}$, the earlier inverse-differential rule gives

$$d(\mathbf{\Omega}^{-1}) = -\mathbf{\Omega}^{-1}(d\mathbf{\Omega})\mathbf{\Omega}^{-1}.$$

Since $\mathbf{X}$ and $\mathbf{y}$ are fixed,

$$\begin{aligned}\mathbf{0} &= \mathbf{X}'d(\boldsymbol{\Omega}^{-1})\widehat{\boldsymbol{\varepsilon}}_G - \mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X}d\widehat{\boldsymbol{\beta}}_G, \\ d\widehat{\boldsymbol{\beta}}_G &= -\mathbf{V}\mathbf{X}'\boldsymbol{\Omega}^{-1}(d\boldsymbol{\Omega})\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{\varepsilon}}_G.\end{aligned}$$

For $d\boldsymbol{\Omega} = c\boldsymbol{\Omega}$, the last expression is

$$-c\mathbf{V}\mathbf{X}'\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{\varepsilon}}_G = \mathbf{0}.$$

Exercise 14.10 (GLS under a nonsingular change of observations). Under Theorem 14.10, let $\mathbf{L} \in \mathbb{R}^{n \times n}$ be nonsingular and define

$$\mathbf{y}_* = \mathbf{L}\mathbf{y}, \quad \mathbf{X}_* = \mathbf{L}\mathbf{X}, \quad \boldsymbol{\Omega}_* = \mathbf{L}\boldsymbol{\Omega}\mathbf{L}'.$$

Prove that GLS computed from $(\mathbf{y}_*, \mathbf{X}_*, \boldsymbol{\Omega}_*)$ has the same coefficient vector and covariance matrix as GLS computed from $(\mathbf{y}, \mathbf{X}, \boldsymbol{\Omega})$.

Solution. Since

$$\Omega_*^{-1} = \mathbf{L}^{-T} \Omega^{-1} \mathbf{L}^{-1},$$

we have

$$\begin{aligned} \mathbf{X}'_* \Omega_*^{-1} \mathbf{X}_* &= \mathbf{X}' \Omega^{-1} \mathbf{X}, \\ \mathbf{X}'_* \Omega_*^{-1} \mathbf{y}_* &= \mathbf{X}' \Omega^{-1} \mathbf{y}. \end{aligned}$$

Both the GLS coefficient formula and $(\mathbf{X}' \Omega^{-1} \mathbf{X})^{-1}$ are therefore unchanged. ■

Exercise 14.11 (Aitken's theorem for linear functions). Under Theorem 14.10, let $\mathbf{A} \in \mathbb{R}^{s \times p}$ be fixed. Prove that

$$\mathbf{A} \hat{\boldsymbol{\beta}}_G$$

is the unique BLUE of $\mathbf{A}\boldsymbol{\beta}$, using the definition of BLUE with $\mathbf{A}\boldsymbol{\beta}$ in place of $\boldsymbol{\beta}$, and has covariance $\mathbf{A}\mathbf{V}\mathbf{A}'$. If $\mathbf{C} \in \mathbb{R}^{s \times n}$ and $\mathbf{C}\mathbf{y}$ is any linear unbiased estimator of $\mathbf{A}\boldsymbol{\beta}$, prove

$$\text{cov}(\mathbf{C}\mathbf{y}) - \mathbf{A}\mathbf{V}\mathbf{A}' = (\mathbf{C} - \mathbf{A}\mathbf{G})\Omega(\mathbf{C} - \mathbf{A}\mathbf{G})' \succeq \mathbf{0}.$$

Solution. Unbiasedness is equivalent to $\mathbf{CX} = \mathbf{A}$. Since $\mathbf{GX} = \mathbf{I}_p$, the candidate $\mathbf{AGy}$ is unbiased and has covariance

$$\mathbf{AG}\Omega\mathbf{G}'\mathbf{A}' = \mathbf{AVA}'.$$

Put $\mathbf{D} = \mathbf{C} - \mathbf{AG}$. Then $\mathbf{DX} = \mathbf{0}$ and

$$\mathbf{AG}\Omega\mathbf{D}' = \mathbf{AVX}'\mathbf{D}' = \mathbf{0}.$$

Expanding $(\mathbf{AG} + \mathbf{D})\Omega(\mathbf{AG} + \mathbf{D})'$ proves the identity. Positive definiteness of Ω gives $\mathbf{D}\Omega\mathbf{D}' = \mathbf{0} \iff \mathbf{D} = \mathbf{0}$. If $\mathbf{Cy}$ were also BLUE, comparison with $\mathbf{AGy}$ would give

$$-\mathbf{D}\Omega\mathbf{D}' = \text{cov}(\mathbf{AGy}) - \text{cov}(\mathbf{Cy}) \succeq \mathbf{0}.$$

Together with $\mathbf{D}\Omega\mathbf{D}' \succeq \mathbf{0}$, this implies $\mathbf{D} = \mathbf{0}$; hence the BLUE is unique. ■

Exercise 14.12 (Expectation of the scaled scatter matrix). Let $\mathbf{y}_1, \dots, \mathbf{y}_n \in \mathbb{R}^m$ satisfy

$$\mathbb{E}[\mathbf{y}_i] = \boldsymbol{\mu}, \quad \text{cov}(\mathbf{y}_i) = \boldsymbol{\Omega}, \quad \mathbb{E}[(\mathbf{y}_i - \boldsymbol{\mu})(\mathbf{y}_j - \boldsymbol{\mu})'] = \mathbf{0} \quad (i \neq j).$$

For

$$\bar{\mathbf{y}} = \frac{1}{n} \sum_i \mathbf{y}_i, \quad \widehat{\boldsymbol{\Omega}} = \frac{1}{n} \sum_i (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})',$$

prove

$$\mathbb{E}[\widehat{\boldsymbol{\Omega}}] = \frac{n-1}{n} \boldsymbol{\Omega}.$$

Deduce

$$\mathbb{E}[\widehat{\boldsymbol{\Omega}}] - \boldsymbol{\Omega} = -\frac{1}{n} \boldsymbol{\Omega}.$$

Use only the displayed moments.

Origin: Course formulation from A&M Ex. 13.61

Solution. Put $\mathbf{z}_i = \mathbf{y}_i - \boldsymbol{\mu}$ and $\bar{\mathbf{z}} = n^{-1} \sum_i \mathbf{z}_i$. The stated moments give

$$\mathbb{E}[\bar{\mathbf{z}}\bar{\mathbf{z}}'] = \frac{1}{n^2} \sum_{i,j} \mathbb{E}[\mathbf{z}_i \mathbf{z}_j'] = \frac{1}{n} \boldsymbol{\Omega}.$$

The scatter identity yields

$$\sum_i (\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})' = \sum_i \mathbf{z}_i \mathbf{z}_i' - n \bar{\mathbf{z}} \bar{\mathbf{z}}'.$$

Taking expectations and dividing by n gives

$$\mathbb{E}[\widehat{\boldsymbol{\Omega}}] = \frac{1}{n} \{n \boldsymbol{\Omega} - \boldsymbol{\Omega}\} = \frac{n-1}{n} \boldsymbol{\Omega}.$$

Subtracting $\boldsymbol{\Omega}$ gives $\mathbb{E}[\widehat{\boldsymbol{\Omega}}] - \boldsymbol{\Omega} = -\boldsymbol{\Omega}/n$. 

Exercise 14.13 (Inverse information in symmetric coordinates). Let $\mathbf{D}_m^+$ be the Moore–Penrose inverse of the duplication matrix. Let $\boldsymbol{\Omega} = \boldsymbol{\Omega}' \in \mathbb{R}^{m \times m}$ with $\boldsymbol{\Omega} \succ \mathbf{0}$. Prove

$$\{\mathbf{D}_m'(\boldsymbol{\Omega}^{-1} \otimes \boldsymbol{\Omega}^{-1})\mathbf{D}_m\}^{-1} = \mathbf{D}_m^+(\boldsymbol{\Omega} \otimes \boldsymbol{\Omega})(\mathbf{D}_m^+)'.$$

Deduce from Theorem 14.15 that

$$n\mathcal{I}(\boldsymbol{\theta})^{-1} = \begin{pmatrix} \boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & 2\mathbf{D}_m^+(\boldsymbol{\Omega} \otimes \boldsymbol{\Omega})(\mathbf{D}_m^+)' \end{pmatrix}.$$

Origin: Adapted from A&M Ex. 13.65(b)

Solution. Put

$$\mathbf{M} = \mathbf{D}_m\mathbf{D}_m^+, \quad \mathbf{K} = \boldsymbol{\Omega}^{-1}, \quad \mathbf{A} = \mathbf{D}_m'(\mathbf{K} \otimes \mathbf{K})\mathbf{D}_m, \quad \mathbf{B} = \mathbf{D}_m^+(\boldsymbol{\Omega} \otimes \boldsymbol{\Omega})(\mathbf{D}_m^+)'.$$

The earlier duplication-matrix identities make $\mathbf{M}$ the projector onto vectorised symmetric matrices. If $\mathbf{H} = \mathbf{H}'$, then

$$(\boldsymbol{\Omega} \otimes \boldsymbol{\Omega}) \text{vec } \mathbf{H} = \text{vec}(\boldsymbol{\Omega}\mathbf{H}\boldsymbol{\Omega}), \quad (\mathbf{K} \otimes \mathbf{K}) \text{vec } \mathbf{H} = \text{vec}(\mathbf{K}\mathbf{H}\mathbf{K}),$$

and both matrices on the right are symmetric. Hence the inserted projectors below act as identities:

$$\begin{aligned}\mathbf{AB} &= \mathbf{D}'_m(\mathbf{K} \otimes \mathbf{K})\mathbf{M}(\mathbf{\Omega} \otimes \mathbf{\Omega})(\mathbf{D}_m^+)' \\ &= \mathbf{D}'_m(\mathbf{D}_m^+)' = \mathbf{I}, \\ \mathbf{BA} &= \mathbf{D}_m^+(\mathbf{\Omega} \otimes \mathbf{\Omega})\mathbf{M}(\mathbf{K} \otimes \mathbf{K})\mathbf{D}_m \\ &= \mathbf{D}_m^+\mathbf{D}_m = \mathbf{I}.\end{aligned}$$

Thus $\mathbf{B} = \mathbf{A}^{-1}$. Inverting the two diagonal blocks of Theorem 14.15 gives the stated formula. ■

Exercise 14.14 (Coefficient influence and leave-one-out residuals). Under Theorem 14.17, suppose every row deletion leaves full column rank. Prove

$$(\widehat{\boldsymbol{\beta}}_{-i} - \widehat{\boldsymbol{\beta}})' \mathbf{Q} (\widehat{\boldsymbol{\beta}}_{-i} - \widehat{\boldsymbol{\beta}}) = \frac{\widehat{e}_i^2 h_i}{(1 - h_i)^2}.$$

If $\widehat{e}_{(i)} = y_i - \tilde{\mathbf{x}}_i' \widehat{\boldsymbol{\beta}}_{-i}$, prove the exact identity

$$\sum_{i=1}^n \widehat{e}_{(i)}^2 = \sum_{i=1}^n \left(\frac{\widehat{e}_i}{1 - h_i} \right)^2.$$

Origin: Course formulation from H Cor. 18.2.10; CLM Ex. B1

Solution. Theorem 14.17 gives

$$\widehat{\boldsymbol{\beta}}_{-i} - \widehat{\boldsymbol{\beta}} = -\frac{\mathbf{Q}^{-1} \tilde{\mathbf{x}}_i \widehat{e}_i}{1 - h_i}.$$

Therefore

$$\begin{aligned} (\widehat{\boldsymbol{\beta}}_{-i} - \widehat{\boldsymbol{\beta}})' \mathbf{Q} (\widehat{\boldsymbol{\beta}}_{-i} - \widehat{\boldsymbol{\beta}}) &= \frac{\widehat{e}_i^2}{(1 - h_i)^2} \tilde{\mathbf{x}}_i' \mathbf{Q}^{-1} \tilde{\mathbf{x}}_i \\ &= \frac{\widehat{e}_i^2 h_i}{(1 - h_i)^2}. \end{aligned}$$

The held-out residual identity in that theorem gives $\widehat{e}_{(i)} = \widehat{e}_i/(1 - h_i)$; squaring and summing proves the second formula. ■

Exercise 14.15 (Geometry of restricted covariance). Let $\mathbf{V} = \mathbf{V}' \in \mathbb{R}^{p \times p}$ with $\mathbf{V} \succ \mathbf{O}$, let $1 \leq q \leq p$, let $\mathbf{R} \in \mathbb{R}^{q \times p}$ have rank q , and put

$$\mathbf{S} = \mathbf{RVR}', \quad \mathbf{C}_R = \mathbf{V} - \mathbf{VR}'\mathbf{S}^{-1}\mathbf{RV}.$$

Prove

$$\mathbf{C}_R \succeq \mathbf{O}, \quad \text{rank } \mathbf{C}_R = p - q, \quad \text{null } \mathbf{C}_R = \text{col } \mathbf{R}', \quad \mathbf{RC}_R = \mathbf{O}.$$

Origin: Course formulation from H §19.7; CLM §10

Solution. Let

$$\mathbf{A} = \mathbf{RV}^{1/2}, \quad \mathbf{P} = \mathbf{A}'(\mathbf{AA}')^{-1}\mathbf{A}.$$

Then $\text{rank } \mathbf{A} = q$, $\mathbf{P} = \mathbf{P}' = \mathbf{P}^2$, and $\text{rank } \mathbf{P} = q$. Moreover,

$$\mathbf{C}_R = \mathbf{V}^{1/2}(\mathbf{I}_p - \mathbf{P})\mathbf{V}^{1/2} \succeq \mathbf{O}.$$

Nonsingularity of $\mathbf{V}^{1/2}$ gives

$$\text{rank } \mathbf{C}_R = \text{rank}(\mathbf{I}_p - \mathbf{P}) = p - q.$$

Direct multiplication gives

$$\mathbf{C}_R \mathbf{R}' = \mathbf{V} \mathbf{R}' - \mathbf{V} \mathbf{R}' \mathbf{S}^{-1} \mathbf{S} = \mathbf{O}.$$

Thus $\text{col } \mathbf{R}' \subseteq \text{null } \mathbf{C}_R$. Both spaces have dimension q , so they are equal. Since $\mathbf{C}_R$ is symmetric, $\mathbf{R} \mathbf{C}_R = (\mathbf{C}_R \mathbf{R}')' = \mathbf{O}$. 